

UNIVERSITY OF GREENWICH  
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Integrated Study and Research  
Repository with Flashcard-Based  
Learning

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# ABSTRACT

This project investigates the design and implementation of an integrated web-based system that combines a research repository with flashcard-based learning to enhance how students organise academic resources and retain key information. The study is based on existing research that identifies difficulties such as information overload, fragmented digital learning environments and limited effectiveness of isolated revision tools. To address these issues, a prototype system was developed using PHP, HTML and a MySQL relational database, following an iterative Agile informed approach. The system allows for structured resource management, keyword-based retrieval and the creation of flashcards that are contextually linked to their original sources, supporting retrieval-based learning methods. Evaluation was conducted through white box, black box, integration and usability testing to assess functionality, reliability and user interaction. The results indicate that integrating repository and revision functionalities within a single platform improves workflow consistency and supports more meaningful engagement with academic content. While the prototype indicates the practicality of this approach, limitations remain in areas such as automation and scalability, providing guidance for further development.

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  1. (Arden et al., 2024)
  2. (Carpenter, Pan and Butler, 2022)
  3. (Liu and Khalil, 2023)
  4. (Sillence et al., 2022)
  5. (Varkey Nadakkavukaran Santhosh et al., 2024)
- In other ways, as described below: – YES
  1. For troubleshooting code
  2. For finding some sources when I couldn't

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# 1 INTRODUCTION

The growing dependence on digital tools in higher education has fundamentally altered how students access, arrange, and interact with academic content. While digital platforms offer increased flexibility and accessibility, they also present new obstacles, notably in terms of information overload and fragmented study patterns. Students commonly use various technologies to manage their learning, such as cloud storage systems for resource organisation and different applications for assessment. This disconnected approach can decrease efficiency, raise cognitive load, and restrict the efficacy of study tactics.

This project addresses these problems by emphasising the integration of academic resource management and active learning techniques into a unified system. Existing technologies frequently allow either organisation or revision, but rarely do both in a single setting. As a result, students must manually move material across platforms, which disrupts learning continuity and decreases participation. The problem domain is thus both relevant and meaningful, since it mirrors a prevalent challenge encountered by students in current digital learning contexts.

The aim of this project is to create a web-based application that integrates a research library with a flashcard-based learning system. The suggested approach enables users to save and arrange academic resources while also creating and evaluating flashcards directly related to those topics. This integrated approach aims to eliminate fragmentation, increase workflow efficiency, and enhance evidence-based learning tactics like active recall.

To achieve this aim, the project addresses three significant problems. These include how to properly arrange and maintain academic materials in a database, how to generate relevant revision material from stored content, and how to create an intuitive and scalable system. The research also investigates how contextual linkage between flashcards and source material might improve understanding and retention while keeping learning relevant to its original context.

The project's scope includes the design, development, and assessment of a functioning prototype. This involves requirements analysis, system design with UML and database modelling, web-based implementation, and system testing to ensure functioning. The project also assesses the system's efficacy in solving the given problem, taking into account both its strengths and limitations.

The technique used for this project is iterative and Agile-inspired, allowing for ongoing development, testing, and modification. This guarantees that the system changes in response to new difficulties

while remaining relevant to user demands. Each stage of the project, from preliminary study to final review, is guided by both academic literature and practical factors.

Overall, this project aims to help design more effective digital learning resources by demonstrating how integrating resource management and active recall may enhance student learning experiences.

## **1.1 Background**

University students are increasingly challenged by dual demands of managing large volumes of academic material and retaining key information for long term use. As higher education shifts to digital learning environments, the volumes of notes have grown substantially through articles, blogs, books and more being available online, this has often led to disorganisation and information overload for higher education students. Sillence et al. (2022) found that while students have found various ways to manage their university-based data there are still many who experience difficulties with information overload. Similarly, Arden et al. (2024) highlights how students adapt to digital note-taking practices in lectures but also face different challenges trying to effectively integrate them into the learning routines. These findings suggest that there is a need for more effective systems that support both resource management and effective study practices.

Also, research in educational psychology consistently demonstrates that practices based on spaced repetition are among the most successful practices for improving long term retention. Varkey et al. (2024) showed that a mobile flashcard application consisting of spaced repetition significantly enhanced dental students' long-term knowledge compared to more traditional methods. Similarly, Lu et al. (2021) reported that the use of the flashcard application Anki was effective for better learning efficiency and stronger in-depth medical knowledge.

This project attempts to help higher education students overcome two typical challenges - knowledge retention and information overload. It will accomplish this by creating a web-based program that combines a research repository and a flashcard learning system. Academic materials saved in the database may be accessed directly via the platform, and each source will be connected to automatically created flashcards that emphasise key points or subjects. When students examine these flashcards, selecting a certain point directs them to the matching section of the original source. This study aims to improve the efficiency and engagement of students in organising, accessing, and retaining academic material by integrating systematic resource management with active recall.

## **1.2 Aims and Objectives**

The aim of this project is to develop a web-based application that combines a research library with flashcards to assist students in organising academic resources, extracting key concepts, and

practicing active recall. Therefore, improving information retention, reducing overload, and encouraging better study habits.

1.2.1 Research into existing research repository tools and flashcard systems, evaluating their functionality and limitations through academic sources. [5.0]

1.2.2 Write a literature review on the importance of information management systems and active recall methods in higher education, with clear analysis of assumptions and limitations. [12.0]

1.2.3 Research functional and non-functional requirements for an integrated repository and flashcard system, based on insights from the literature review and user expectations. [3.0]

1.2.4 Verify requirements by designing and distributing a questionnaire or short survey to students on their study habits and tool preferences. [5.0]

1.2.5 Create a rich picture diagram showing students, their study workflows, and how repository and flashcard features interact. [3.0]

1.2.6 Create use-case diagrams illustrating system functionalities such as uploading resources, generating flashcards, and reviewing content. [3.0]

1.2.7 Create a conceptual ERD to outline the entities and their relationships. [3.0]

1.2.8 Create a physical ERD diagram specifying attributes, keys, and database schema structure. [6.0]

1.2.9 Design possible user interfaces using tools such as Figma or Axure RP. [7.0]

1.2.10 Implement the database schema by creating tables and relationships in MySQL or MongoDB. [10.0]

1.2.11 Develop the main user interface using a web framework using PHP and Html. [13.0]

1.2.12 Implement the resource upload and organisation features, allowing users to add, tag, and search academic sources. [8.0]

1.2.13 Implement flashcard generation linked to stored resources, displaying key concepts or topics. [13.0]

1.2.14 Create functionality where clicking a flashcard topic redirects users to the relevant section in the source. [8.0]

1.2.15 Implement user registration and login functionality with secure authentication. [8.0]

1.2.16 Add an admin panel to manage users, resources, and system data. [8.0]

1.2.17 Usability testing to observe how easily students navigate the repository and flashcard features. [5.0]

1.2.18 Black box testing to ensure system functions work as expected for users unfamiliar with the code. [5.0]

1.2.19 White box testing to verify system logic and database interactions for flashcard generation and resource retrieval. [5.0]

1.2.20 Evaluate whether the system meets the initial functional and non-functional requirements. [5.0]

1.2.21 Assess the system's usefulness in reducing information overload and improving knowledge retention through small-scale user feedback. [5.0]

1.2.22 Discuss limitations, challenges, and potential improvements for future development. [5.0]

### **1.3 Methodology**

The Agile methodology is adopted for this project because it supports a user-friendly approach and an extremely flexible approach to system development, which is essential for building the learning platform that integrates a research repository and flashcards. The process enables the project to move through organised but adaptive stages of design, implementation, testing, and evaluation, guaranteeing that each system component is constantly modified in response to user feedback.

During the design stage, Agile facilitates the construction of user-centered interface prototypes, such as the repository dashboard, flashcard viewer, and source-linking functions, which are jointly built to meet student demands. The implementation and development stages are divided into small, manageable sprints, allowing for incremental improvements and rapid adjustments depending on testing findings. Although this project is being created independently, Scrum concepts will be modified for solo usage by considering each sprint as a concentrated development period with well-defined goals, such as design, implementation, and testing. Progress will be tracked via self-review at the conclusion of each sprint, allowing for reflection, refinement, and job prioritisation in the following phase.

Frequent testing ensures that the interface stays intuitive and that data retrieval and flashcard creation work properly. Finally, the evaluation step compares the project outputs to the original objectives, focusing on usability, functionality, and improved knowledge retention.

According to Santos et al. (2024), Agile practices such as Scrum use adaptability, collaboration and SMART project outcomes within academic environments. Similarly, Friess (2023) found that groups using Scrum achieved higher efficiency through better communication and better performance scores compared to traditional approaches. These findings support the use of agile for this project, as it allows for continued refinement to ensure that the final product meets students' needs for information organisation and long-term knowledge retention.

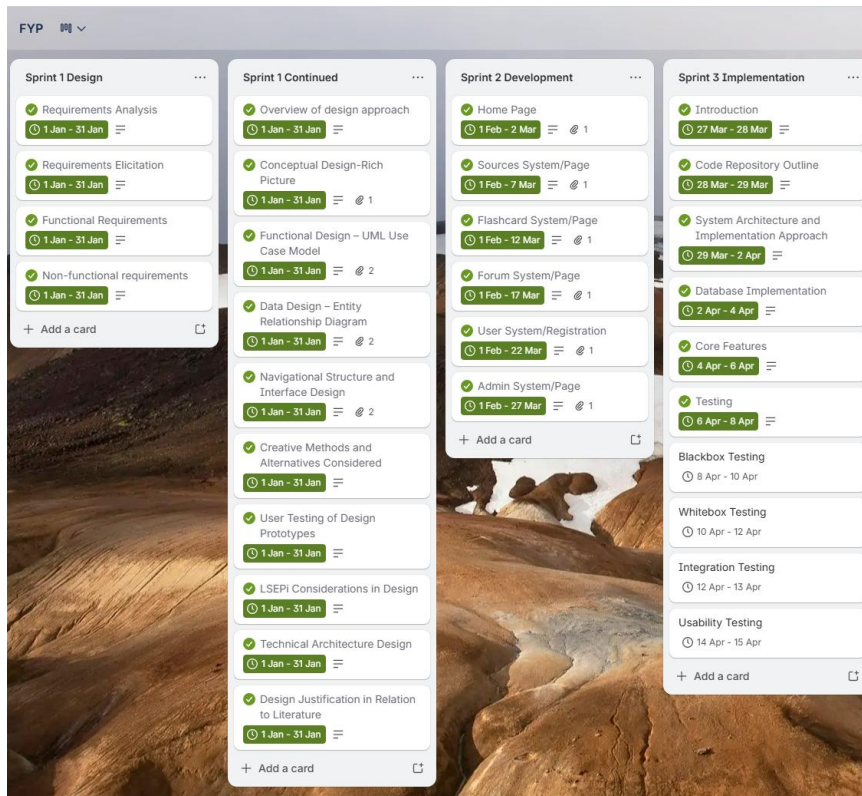


Figure 1

The project is using agile methodology, with tasks arranged into iterative sprints using a Trello board, as seen in Figure 1. Each sprint focuses on a certain stage of development, such as design, core feature development and implementation. Tasks were developed and recorded to represent essential system capabilities such as user authentication, resource management, flashcard integration and testing activities, making progress to be structured and clearly defined.

## 1.4 Report Roadmap

This report is designed to guide the reader through the whole development process. This project's key objective is to create and develop a functional prototype that targets a real-world problem which is the difficulty of structuring academic information and retaining key information. Following the introduction, Chapter 2 provides a literature overview of existing research on digital learning technologies, information management, and flashcard-based learning. Chapter 3 gives a comprehensive assessment of existing systems, highlighting their strengths and limitations. Chapter 4 discusses requirements analysis and system design, including modelling methodologies and interface design. Chapter 5 describes how the system was implemented, and Chapter 6 offers the prototype's findings and evaluation. Chapter 7 examines legal, social, ethical, and professional aspects, and the project finishes with a summary of the results and recommendations for further work.

The project begins with a research-informed phase, which includes a methodical evaluation of academic literature digital resource management, flashcard based learning and spaced repetition. This stage identifies frequent student challenges, such as information overload, inefficient material organisation and insufficient long-term retainment of key information. The findings of the literature review immediately affect the goal and direction of the project, ensuring that the proposed solution is based on current research rather than entirely subjective assumptions.

Following the literature review, the project will use an iterative and continuous development method, rather than attempting to design and build the full system in one step. Initial iterations concentrate on establishing key features, such as user authentication, fundamental repository administration and manual flashcard production. Subsequent production expands on this foundation by offering more complex capabilities like greater organising, filtering and the ability to link flashcards to specific academic sources.

Prototyping is a key component of the selected agile methodology. Low and mid fidelity designs and prototypes are used to test layout, navigation and user interface, allowing the design concepts to be tested prior to final implementation. As the project advances through revisions, these prototypes transform into working versions of the system. This incremental prototyping approach encourages continued testing and improvement, allowing usability concerns to be detected and resolved early in the development cycle.

The implementation phase focuses on creating a web-based database driven prototype that makes use of relevant data storage and system logic technologies. The usefulness of technology for handling structured academic data is impacted by existing data, allowing the project to focus on system designs and functionality rather than spending too much time learning unfamiliar tools.

Evaluation and reflection are used throughout the project's lifespan. The system is constantly evaluated against the established functional and non-functional requirements to see whether it accomplishes its intended goals. Emphasis is placed on usability, resource organisational efficacy, and the system's capacity to facilitate learning and revision. Reflection on both technical and design decision informs the future development and gives critical assessment of the final prototype's strengths and limitations.

Overall, this method promotes the creation of a realistic, research-based prototype while allowing for adequate flexibility to respond to problems and altering understanding during the project.

## 2 LITERATURE REVIEW

### 2.1 Introduction

The literature review will be completed using a structured search approach to identify relevant academic resource management, and flashcard-based learning. The main aim of the literature search was to establish a strong theoretical foundation for the project and guarantee that the proposed system is formed by existing research rather than assumptions.

Google scholar was the primary source for identifying literature, due to its extensive coverage of peer reviewed journal articles, conference papers, and academic publications across computing, education and information systems. In addition to Google Scholar, academic publisher platforms and digital libraries commonly used within computing and educational technology research were consulted, including sources that host peer-reviewed journals and conference proceedings. These platforms were selected because they provide access to credible, research-based material appropriate at the undergraduate level.

Search terms were chosen to represent the project core themes and adjusted iteratively as the scope of the study became clearer. Keywords and phrases included combinations of terms such as digital study tools, academic resource organisation, student information overload, flashcard learning, retrieval practice, spaced repetition and technology-supported learning. Boolean operators were used where appropriate to restrict results and improve relevance. This iterative refinement ensured that the search process focussed on material directly relevant to the projects aims and objectives.

To maintain relevance and academic quality, inclusion and exclusion criteria were applied during the selection process. The priority was peer reviewed journal articles and conference papers published within the last five years to represent the current research and technological practices. However, a small number of older studies were cited because they provided foundational theoretical concepts relevant to learning and memory. Studies prioritised on higher education environments, whereas non-academic sources, opinion pieces and articles lacking empirical or theoretical foundation were removed.

Once identified the relevant material was evaluated and organised thematically and chronologically. Reference management software was used to store sources, manage citations and generate a bibliography in the Harvard referencing style format. This approach supported consistency, reduced referencing errors and enabled efficient management of a growing body of literature.

Overall, this methodical approach to literature searching guaranteed that the evaluation was thorough, relevant and academically rigorous, providing a strong foundation for identifying the problem and informing subsequent requirements and design of the proposed system.

## 2.2 Identifying the problem

The literature identifies several interrelated issues that university students encounter as learning where more learning takes place in a digital environment. According to Sillence et al. (2022), one of the most significant challenges encountered is students' difficulty in organising and managing large amounts of academic information. As digital technologies have become significantly more important in higher education, students are required to engage with large volumes of lecture material, academic papers, online resources and personal notes. While these technologies offer improved access to knowledge, research indicates that they have also contributed to information overload and fragmented study habits. Sillence et al. (2022) shows how students accumulate academic data across multiple platforms frequently without a clear organisational strategy. Their findings indicate that students frequently have trouble identifying crucial material when rewriting, resulting in inefficiencies and frustration. The study emphasises that although students make considerable use of digital technologies, these tools do not always support effective long-term retrieval of information. This is especially problematic during assessment periods, where the ability to acquire and synthesis key information is crucial.

In addition to organisational challenges, the literature indicates that digital learning environments often rely on a fragmented ecosystem of technologies. Bond et al. (2018), in their study of digital transformation in higher education, highlights that learning materials are usually implemented in isolation rather than part of an integrated system. Students may use one platform for storing key information, another for storing resources and a separate tool for revision or testing. This framework can impair learning continuity and increase cognitive load, and students must constantly switch contexts and mentally link knowledge across systems.

The separation of information storage and active learning is a recurring concern in the literature. While repositories and learning management systems facilitate information distribution and storage, they do not promote deep learning or retention. Bond et al. (2018), argues that many digital learning tools emphasise access and delivery above meaningful engagement, which can lead to surface level learning habits. This discrepancy implies that effective learning requires not only access to information but also structured opportunities for active engagement with that information.

Research into learning strategies provides further insight into this problem. In their systematic evaluation of interleaving as a concept learning strategy, Firth et al. (2021) emphasises the need for spreading and diversifying learning sessions to promote deeper knowledge retention and comprehension. Their findings indicate that organised learning strategies such as interleaving and spacing, can significantly increase learning outcomes when compared to massed or repetitive study. However, the review also shows that such strategies are not consistently embedded within everyday study practices or supported by frequently used digital tools.

Trumble et al. (2023), emphasises the advantages of retrieval practice and distributed learning systems. Their thorough research shows strategies that requires learners to actively recall information for better long-term retention and knowledge transfer. Despite this strong evidence base, Trumble et



al. (2023) observes that retrieval-based strategies are often implemented from students overall learning processes.

A further issue identified in the literature is the absence of an explicit relationship between revision material and its academic source. Firth et al. (2021) and Trumble et al. (2023) argue that learning procedures are more effective when students can relate recalled material to its original context.

However, most digital study tools lack systematic assistance for integrating revision activities to specific lecture notes, reading or research papers. This increases cognitive demands on students and may discourage long term engagement with evidence-based learning practices.

When taken together, these studies demonstrate a clear problem space. Students are facing increasing information overload by relying on fragmented digital tools and frequently lack structured assistance for incorporating good learning habits into their study routines (Sillence et al., 2022; Bond et al., 2018).

Although research strongly supports retrieval practice, spacing and interleaving, existing systems rarely integrate these approaches with effective resource management or contextual linking (Firth et al., 2021; Trumble et al., 2023).

This gap directly influences the scope of the planned project. According to the research, there is a need for an integrated system that supports both academic resource organisation and the implementation of evidence-based learning practices in one educational environment. By addressing issues of fragmentation, information overload, and weak contextual connections between learning materials and revision activities, the project hopes to directly address the identified challenges while also laying the groundwork for the requirements and design decisions discussed in subsequent chapters.

## **2.3 Repository systems in higher education: Benefits and Limitations**

Digital repository systems are a crucial component of modern higher education, providing students with platforms for storing, organising, and accessing academic content. These technologies include learning management systems, cloud-based storage, and digital note repositories, all of which have grown in importance as institutions transition to mixed and wholly digital learning settings. The research consistently emphasises that repository systems provide considerable benefits in terms of accessibility, flexibility, and scalability, allowing students to interact with learning resources across devices and environments (Bond et al., 2018).

One of the key advantages of repository systems is their ability to help organise and centralise information. Arden et al. (2024) notes that students are increasingly relying on digital repositories to help organise academic content, readings and personal notes. These systems allow students to efficiently store key information during lectures and support the transition from paper-based study. From an institutional perspective, repositories also provide consistency in content distribution and reduce the risk of information loss.

While repository systems are good at preserving knowledge, the literature indicates that they frequently serve as passive containers rather than active learning environments. Sillence et al. (2022) emphasises that students frequently accumulate large amounts of academic data in digital repositories without clear organisational systems. Their findings indicate that this accumulation can lead to information overload, making it harder for students to identify relevant material while revising. As repositories expand over time, their value may decrease, especially if students rely on folder structures or inconsistent naming conventions.

A further limitation identified in the literature is lack of support for meaningful engagement with stored content. Bond et al. (2018) argues that many digital repositories prioritise access and efficiency over effectiveness. Repository systems are often designed to distribute content rather than facilitate how students digest, revisit or apply the content. As a result, students may engage in surface level study habits like reviewing notes rather than deeper forms of learning that promote long term retention.

Research on study habits reinforces this concern. Arden et al. (2024) argues that while students change their note-taking practices to digital contexts, these changes do not necessarily result in better learning results. Although digital notes and repositories can indicate quick information collection, the learning value of recorded content may be restricted if organised opportunities for review and reflection are not provided. This highlights a key issue within repository-based systems which is that they excel at collecting information but often fail to support subsequent stages of learning.

The literature also implies that repository systems are mostly unrelated to evidence-based learning methodologies. According to Trumble et al. (2023) and Firth et al. (2021) strategies such as retrieval practice and spread learning are particularly successful in promoting long-term retention. However, these strategies are rarely embedded into repository platforms. Instead, students often rely on separate digital tools for revision activities, resulting in a disconnect between where material is kept and how it is learned.

This separation can reduce the effectiveness of both systems. Trumble et al. (2023) argues that retrieval-based learning is most successful when students effortlessly connect recalled knowledge to its original academic context. When repositories and revision tools operate independently, students must manually re-establish these connections, which increases cognitive effort and reduces efficiency. Sillence et al. (2022) also states that students frequently struggle to maintain consistent organisational structures across numerous digital platforms, leading to fragmentation.

Synthesising these findings reveals that repository systems offer clear benefits in terms of access, storage and flexibility, but they also have substantial limitations in facilitating good learning habits. While repositories effectively manage academic content, they frequently fail to encourage active participation, structured revision or contextual linkage between materials. According to the study, these constraints are caused by system designs that prioritise storage over learning, rather than a lack of student motivation (Bond et al., 2018; Arden et al., 2024).

This analysis offers a compelling argument for the planned investigation. Rather of replacing repository systems, the literature encourages the creation of improved repository systems that incorporate learning-oriented features. Many of the issues raised might be addressed by an integrated repository that promotes organisation while also allowing for retrieval-based learning and contextual linking. These findings immediately inform the planned project and lay the foundation for the requirements and design considerations described in following chapters.

## **2.4 Flashcard systems in higher education: Benefits and Limitations**

Flashcards remain one of the most prominent revision tools in higher education because they correspond well with evidence-based learning principles, especially retrieval practice and separated practice. Rather than providing passive review, flashcards encourage learners to develop an answer, which strengthens memory and improves future performance. This is reflected in the health-professions education literature, where retrieval and distributed practice are frequently linked with greater academic achievements with less active study approaches. (Trumble et al., 2023)

One significant advantage of flashcards is that they provide a low-barrier method for operationalising retrieval learning. Students can convert lecture topics into brief prompts that can be practiced repeatedly, making flashcards useful for high-volume material and time-sensitive revision. According to Lu, Farhat and Beck Dallaghan (2021) research on Anki-based learning, using a spaced-repetition flashcard system is associated with better perceived retention and assessment outcomes in medical education environments, supporting the argument that flashcards can serve as an efficient supplement to formal teaching. Furthermore, curriculum-linked electronic flashcards have been demonstrated to be viable and well-received in medical programs, with usage correlated with outcomes in some circumstances, demonstrating that students interact with flashcards when they are associated with course requirements. (Sun et al., 2021)

Another advantage of using flashcards digitally is that they allow for spaced repetition and effective scheduling. According to Carpenter, Pan and Butler (2022) research evaluations, spacing and retrieval are effective methods, but they are generally underutilised unless students employ tools that manage repetition over time. Flashcard systems that provide information at increasing intervals assist in transferring this research into habitual practice, particularly when students are handling numerous modules and require a constant review routine. In practice, this matters because students frequently fail to self-regulate revision without guidance, flashcard scheduling reduces the planning effort by determining what to study next based on past performance.

Flashcards can also be adaptable in terms of question structure, and literature indicates that format selection influences how much learning is absorbed. Prompts might need simple answers, extended

explanations, or recognition-based selection; thus, retrieval practice involves more than just one exercise. Comparing different retrieval formats demonstrates that the design of retrieval questions may influence retention outcomes, demonstrating that the instructional value of flashcards is determined not just by their use but also by how they are structured. This is crucial for the project since it includes assistance for developing high-quality cards rather of presuming that any flashcard would elicit great learning. (van Wijk et al., 2024)

Despite these benefits, the literature identifies significant limitations. According to Carpenter, Pan and Butler (2022) one limitation is that flashcards may encourage fragmented and decontextualised learning by reducing complicated topics to specific information. While retrieval exercise improves memory for the studied material, flashcards may not automatically assist conceptual comprehension unless prompts are constructed to encourage explanation, application, or integration. Reviews of spacing and retrieval stress that these methods are effective, but their effectiveness is dependent on the alignment of the learning aim. In other words, flashcards are effective for recall but may be ineffective for higher-order learning unless well designed.

A second limitation is the time and skills needed for developing useful flashcards. According to Lu, Farhat and Beck Dallaghan (2021) students typically develop decks hastily, which might result in poorly written prompts, ambiguous answers, or cards that are too simple. Where studies show beneficial results, students often utilise flashcards consistently and in ways that correspond to course content, this indicates that advantages may be reduced when students generate low-quality cards or neglect review habits. This is a design implication which should allow systems both card storage and quality control.

A third limitation is an inconsistency between assessment and educational structure. Flashcards are most successful when the prompts correspond to what students are expected to know and do in exams. According to Sun et al. (2021) research on curriculum-related flashcards, consumption and effectiveness increases when flashcards are explicitly linked to taught material and learning outcomes. Without such alignment, flashcards may become a study tool that competes with rather than reinforces course information, reducing motivation and long-term use.

Finally, although retrieval practice and spaced repetition are widely recommended, research shows that children do not always use these methods efficiently without guidance and feedback. According to Carpenter, Pan and Butler (2022) reviews, even when learners are aware of beneficial tactics, they frequently underuse them, indicating a gap between research and everyday practice that educational systems could help bridge through reminders, scheduling, and performance feedback. This emphasises the necessity for tool design that promotes consistency rather than depending solely on self-discipline. Overall, the research supports flashcards as an effective device for retrieval practice and spaced learning, especially when digital technologies are used to help structure reviews and sustain engagement. At the same time, flashcards have constraints with decontextualisation, card quality, and

alignment with learning aims. These findings support a design approach in which flashcards are integrated into a larger learning workflow, such as linking cards back to the original notes or sources to preserve context, promoting high-quality prompt design, and incorporating spaced scheduling to make evidence-based practice easier to maintain.

## **2.5 Conclusion**

This literature review has identified multiple important insights that will directly guide the proposed project's path. The studied research consistently shows that, while students extensively utilise digital technologies to save and organise academic resources, they frequently suffer with information overload, fragmented systems, and ineffective revision procedures. Repository systems have obvious advantages in terms of accessibility and centralisation, but they usually serve as passive storage places that does not actively encourage learning or long-term retention.

In contrast, the literature on flashcard-based learning shows substantial support for tactics that actively promote memory and learning. In comparison to passive study methods, retrieval practice and spaced repetition have been proven to boost long-term retention (Trumble et al., 2023). Flashcards are thus an effective learning aid when utilised in a structured and purposeful manner. However, the research identifies significant limitations, such as the possibility of surface-level learning and insufficient contextual linkages between revision materials and original academic sources (Firth et al., 2021).

Taken together, the evidence reveals that a major issue is not a lack of good study tools, but rather a lack of synergy among them. Although repository systems and flashcard tools address complimentary parts of the learning process, they are often viewed as different solutions. This division leads to fragmented study routines and increases cognitive demands on students.

This information will be used to the project's next stages. The requirements analysis will concentrate on identifying qualities that facilitate both effective organising of academic resources and active learning via retrieval practice. The design phase will focus on preserving obvious linkages between stored materials and flashcard-based revision information, as well as resolving contextual gaps highlighted in the literature. Finally, the prototype step will investigate how evidence-based learning techniques might be integrated into a repository environment rather than being developed as separate tools.

Overall, this literature analysis offers a clear explanation for the proposed project and serves as a solid research basis for the requirements, design, and prototyping decisions described in following chapters.

## 3 PRODUCT REVIEW

### 3.1 Introduction

The product review looks at existing digital tools for academic resource management and revision. It assesses their key features, strengths, and limits in promoting student learning. They are scored in section 3.3 using a 1 to 5 metric where 1 means very poor and 5 means excellent. This study serves as a basis for identifying gaps that will guide the design of the proposed system.

### 3.2 Products

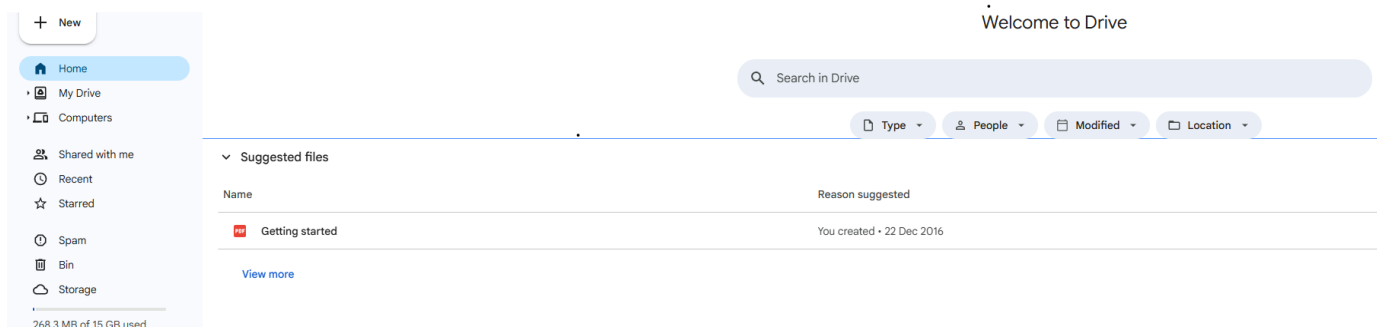


Figure 2

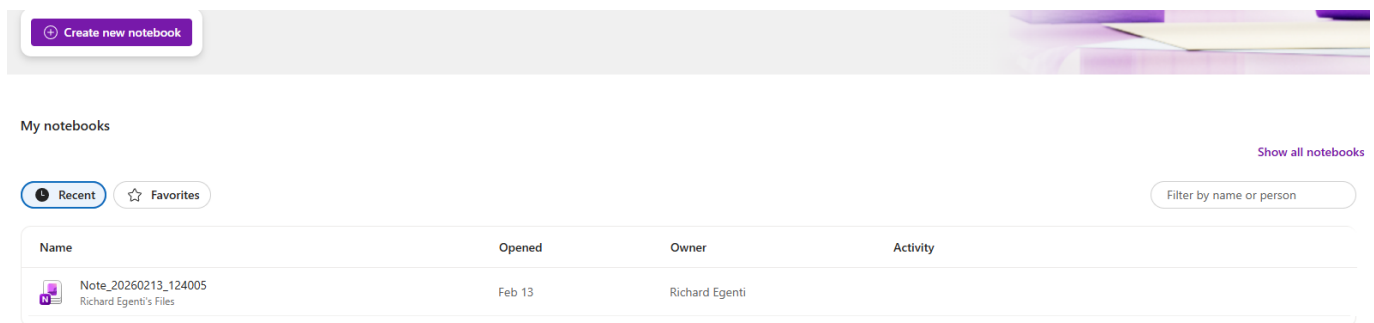


Figure 3

Figure 2 and Figure 3 are both well-known repository systems that are used by students to store any academic material that they have learnt or are going use later to revise. Figure 2 is Google Drive and Figure 3 is Microsoft OneNote.

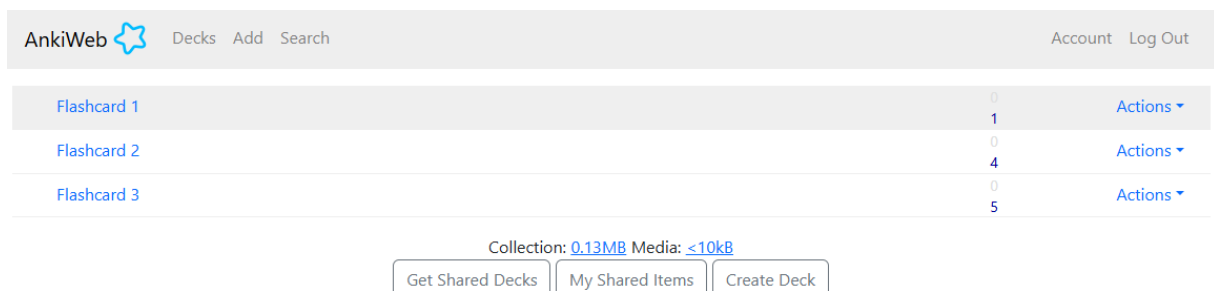


Figure 4

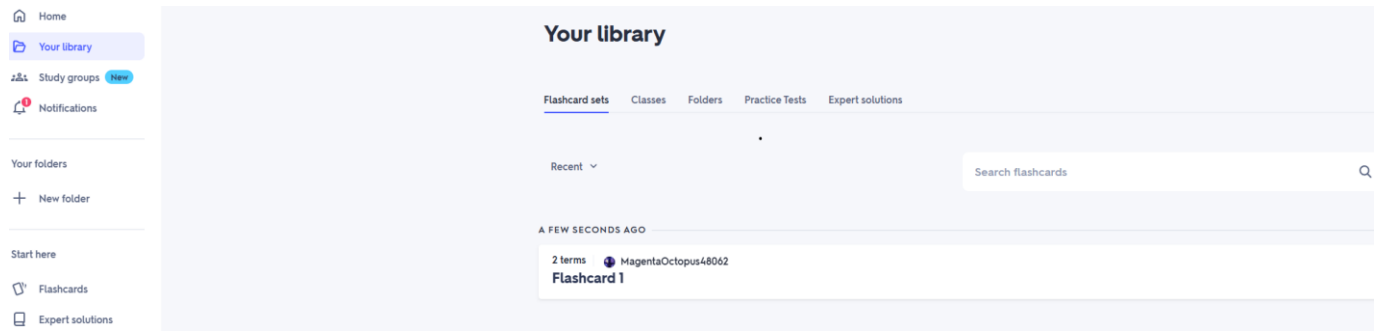


Figure 5

Figure 4 and Figure 5 are both digital flashcard platforms that are used by students to revise any academic material they have gathered or studied. Figure 4 is Anki and Figure 5 is Quizlet.

### 3.3 Product 1 and 2

| ID | Feature                         | Google Drive<br>(Figure 2)                                                                                                                                 | Microsoft One Note<br>(Figure 3)                                                                                                                                        | Rating<br>Figure 2 | Rating<br>Figure 3 |
|----|---------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------|--------------------|
| 1  | Ease of navigation              | Navigation is clear and folder based. However, as repositories grow larger, locating specific files becomes more difficult due to deep folder hierarchies. | Navigation is structured using notebooks and sections, which helps organisation. However, excessive nesting may reduce efficiency when managing large volumes of notes. | 4                  | 4                  |
| 2  | Organisation and Structure      | Provides strong file organisation using folders and file naming. However, structure depends entirely on user management.                                   | Provides structured organisation using notebooks, sections, and pages, allowing logical grouping of information.                                                        | 4                  | 5                  |
| 3  | Support for Learning Activities | Primarily functions as a storage platform. Does not actively support learning or revision methods.                                                         | Supports notetaking but lacks active recall or revision tools. Learning remains passive.                                                                                | 2                  | 3                  |
| 4  | Contextual Linkage              | Files are stored independently with no learning linkage or revision connection.                                                                            | Notes are connected within notebooks but not linked to revision tools.                                                                                                  | 2                  | 3                  |
| 5  | Cognitive Load                  | Simple interface but managing large repositories increases cognitive load over time.                                                                       | Clear interface but excessive note accumulation may increase complexity.                                                                                                | 3                  | 4                  |
| 6  | User Effort Required            | Requires manual organisation and maintenance of files.                                                                                                     | Requires manual note organisation and review.                                                                                                                           | 3                  | 3                  |



### 3.4 Product 3 and 4

| ID | Feature                         | Anki<br>Figure 4                                                                                                | Quizlet<br>Figure 5                                                                                 | Rating<br>Figure 4 | Rating<br>Figure 5 |
|----|---------------------------------|-----------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|--------------------|--------------------|
| 1  | Ease of navigation              | Navigation is functional but less intuitive for new users. Interface complexity can make initial use difficult. | Navigation is simple and user-friendly. Clear menus and visual layout make features easy to locate. | 3                  | 5                  |
| 2  | Organisation and Structure      | Organises flashcards into decks effectively but lacks broader academic resource management.                     | Flashcards are well organised into sets but lacks integration with academic source materials.       | 4                  | 4                  |
| 3  | Support for Learning Activities | Strong support for active recall and spaced repetition. Encourages effective revision habits.                   | Supports revision through flashcards and quizzes but emphasises recognition over deep recall.       | 5                  | 4                  |
| 4  | Contextual Linkage              | Flashcards exist independently and are not directly linked to academic sources.                                 | Flashcards lack strong linkage to original academic source materials.                               | 3                  | 3                  |
| 5  | Cognitive Load                  | Automated scheduling reduces cognitive load during revision.                                                    | User-friendly interface reduces cognitive effort during revision.                                   | 5                  | 5                  |
| 6  | User Effort Required            | Requires initial setup and manual flashcard creation, but automated revision reduces ongoing effort.            | Easy to create flashcards but maintaining academic context requires manual effort.                  | 4                  | 4                  |

### **3.5 Key findings**

The research shows that although repository platforms like Google Drive and Microsoft OneNote excel in organisation and accessibility, they lack when it comes to active learning and revision. Flashcard platforms, such as Anki and Quizlet, provide significant assistance for retrieval practice and revision but lack contextual interaction with academic source materials. Anki has the best support for active recall but suffers from usability issues, whereas Quizlet has a more intuitive UI but less contextual learning assistance. These findings support the usability gap highlighted in the literature, which requires students to manage and update academic information using various unconnected technologies. The proposed system addresses this issue by combining repository capabilities and flashcard revision within a single platform, therefore lowering cognitive overload and increasing workflow efficiency.

### **3.6 Conclusion**

Overall, the comparison shows that both repository systems and flashcard applications offer valuable but incomplete solutions for students. Repository systems promote organisation and accessibility, whereas flashcard platforms improve active recall and revision. However, neither category properly merges all the functions into a single environment. This fragmentation forces students to use several technologies, increasing cognitive load and decreasing efficiency. These findings justify the development of the proposed integrated system, which attempts to combine organised resource management with effective revision support in a single environment.

## **4 REQUIREMENTS ANALYSIS AND DESIGN**

### **4.1 Introduction**

This section describes the proposed platform's requirements analysis and system design. It translates the theoretical ideas identified from the literature review and product comparison into organised functional and non-functional requirements. These requirements form the foundation for the systems architecture and design interface, which are described with formal modelling approaches such as UML diagrams and Entity Relationship Diagrams. Each design decision is justified in terms of usability principles, system constraints and project objectives ensuring that conceptual research findings are aligned with technical implementation.

### **4.2 Requirements Analysis**

#### **4.2.1 Analysis of Requirements**

The project's requirements analysis is determined by the project's goal, set goals, literature study findings, and insights acquired by examining existing repository and flashcard-based educational tools. The major goal of the project is to develop a web-based tool that combines a research repository with flashcard capabilities to help students organise academic materials, practise active recall, and improve knowledge retention while decreasing cognitive overload. As a result, the system requirements must promote both effective information management and evidence-based learning techniques.

The literature review identified many important issue areas that have a direct impact on the requirements analysis. Studies on student usage of digital tools often highlight difficulties such as information overload, fragmented study processes, and ineffective revising practices. Repository systems were found to be useful for storing and accessing academic content, but they frequently lacked aspects that actively promote learning and retention. Flashcard-based systems, on the other hand, allowed for retrieval practice and spaced learning but were often separated from source information, restricting contextual knowledge. These findings suggest that the system should satisfy both organisational and learning-related criteria in an integrated way.

A review of existing repository and flashcard platforms supplemented the requirements analysis. Popular repository technologies prioritise flexibility and storage capacity but often rely largely on manual organising via folders or tags, which can be difficult to manage over time. Flashcard platforms, while useful for revision, sometimes require users to navigate between systems and manually handle linkages between revision material and original sources. This division raises cognitive burden and creates inefficiencies in the study process. As a result, a key requirement

highlighted is the seamless integration of stored academic materials with flashcard-based revision content.

In addition, user-related concerns are important in requirements analysis. The system is designed for use by students with many different levels of technical knowledge. Therefore, usability and simplicity of navigation are critical. The requirements must thus provide straightforward procedures for adding resources, structuring content, creating flashcards, and evaluating material. Complex structure or excessive manual input might discourage consistent use, reducing the system's learning advantages. This is consistent with the project's aim of decreasing information overload rather than contributing to it.

Security and data management issues were also examined as part of the requirements process. As the system contains user accounts and academic data storage, it is necessary to have secure user authentication, regulated data access, and suitable administration of system roles such as administrators and everyday users. These factors guarantee that the system is appropriate for real-world usage and meets the required criteria for web-based applications.

The project's implementation constraints and development style also influence requirement analysis. The system is a web-based application built using PHP and HTML, with a relational database as backend. This effects needs for data structure, speed, and scalability. The database, for example, must provide efficient interactions between people, resources, and flashcards in order to enable quick retrieval and contextual links. Similarly, the system architecture must accommodate future expansion, allowing new features to be introduced without requiring extensive redesign.

Finally, the requirements analysis is iterative and based on expected user input. This process's feedback is meant to modify and prioritise requirements, ensuring that the system meets actual user demands rather than theoretical assumptions. This user-centred approach guarantees that the criteria are both academically solid and practically applicable.

To summarise, the analysis of requirements uses literature discoveries, product research, user expectations, and technological restrictions to establish what the system must do. This analysis serves as the basis for the functional and non-functional requirements described in the next sections, ensuring that future design and implementation decisions are consistent with the project's overall aim and objectives.

#### 4.2.2 Requirements Elicitation

Requirements elicitation is the process of identifying and refining system requirements to ensure that the system matches user requirements and project objectives. This project's requirements were developed from four main sources, a literature review, product research, project aim and objectives.

The literature review established a theoretical foundation by emphasising significant difficulties such as information overload, fragmented study tools, and inefficient revision methods. Repository systems were shown to enable storage but not active learning, whereas flashcard tools improved retention but lacked contextual linking. These findings resulted in essential needs such as merging repository and flashcards features, enabling retrieval-based learning, and maintaining links between sources and revision content.

Product research supported these findings by identifying strengths and limitations in existing systems. Repository platforms provided accessibility, but they need extensive manual organising and raising cognitive strain. Flashcard technologies enabled active recall, but they were frequently isolated from the source material and needed significant effort to maintain. This resulted in the requirements to be focused on usability, automation, and reducing user effort, such as streamlined workflows for uploading sources and generating flashcards.

Project aims and objectives also influenced requirement development. The purpose of developing a web-based system that combines a research repository and flashcards resulted in functional requirements needs such as resource uploading, tagging, searching, flashcard creation and user authentication. Non-functional requirements like usability, performance, and reliability were influenced by testing and evaluating objectives.

Finally, the elicitation process was iterative. Requirements were refined based on technical constraints to ensure they were both realistic and feasible within the project's scope. In summary, requirements were identified using a structured, multi-source approach, ensuring they are theoretically grounded, user-focused and technically achievable, forming a strong foundation for the system specification.

### 4.2.3 Functional Requirements

| ID   | Functional Requirement   | Description                                                                | Prioritisation |
|------|--------------------------|----------------------------------------------------------------------------|----------------|
| FR1  | User registration        | Allow users to create an account using a unique email and password.        | Must have      |
| FR2  | User Authentication      | Allow registered users to log in and log out securely.                     | Must have      |
| FR3  | User roles               | Support standard user and administrator roles.                             | Must have      |
| FR4  | Resource Upload          | Allow users to request for academic resources to be uploaded.              | Must have      |
| FR5  | Resource organisation    | Allow users to organise resources using tags or modules.                   | Must have      |
| FR6  | Resource search          | Allow users to search for stored resources.                                | Must have      |
| FR7  | Resource viewing         | Allow users to view and access uploaded resources.                         | Must have      |
| FR8  | Flashcard creation       | Allow users to create flashcards manually or from stored resources.        | Must have      |
| FR9  | Flashcard linking        | Link flashcards to their corresponding academic resources.                 | Must have      |
| FR10 | Contextual Navigation    | Allow navigation from a flashcard to the relevant section in the resource. | Should have    |
| FR11 | Flashcard review         | Allow users to review flashcards using active recall.                      | Must have      |
| FR12 | Revision tracking        | Track flashcard review activity for users.                                 | Could have     |
| FR13 | Admin user management    | Allow administrators to manage user accounts.                              | Should have    |
| FR14 | Admin content management | Allow administrators to manage resources and system data.                  | Should have    |
| FR15 | Error handling           | Provide clear system and user feedback messages.                           | Must have      |

#### 4.2.3.1 Functional Requirements Justification

User management requirements FR1-FR3 ensure that the system can distinguish between and support multiple users and administrators. This is essential for protecting user data and enabling administrative oversight, such as managing accounts and system content.

Resource related requirements FR4-FR7 address the core problem of academic material organisation.

The ability to categorise, search and view materials permits users to have a structured repository of

academic content, minimising information fragmentation and improving accessibility during revision periods.

Flashcard related requirements FR8-FR12 support the systems learning oriented functionality, by allowing flashcard to be created and explicitly connected to stored materials, the system maintains contextual links between revision material and original sources. The addition of flashcard review and activity tracking supports active recall and constant engagement with learning materials.

Administrative requirements FR13-FR14 ensures that the system can be managed effectively, especially in scenarios with multiple users or institutional use. Error handling requirements FR15 improves system usability by providing clear feedback and reducing user frustration.

Together, these functional requirements define the systems core characteristics and serve as a solid basis for the design, implementation, and testing stages outlined in the subsequent chapters.

#### **4.2.3.2 Prioritisation Rationale**

The must have requirements represent the core functionality required for the system to meet its primary aim of supporting academic resource organisation and active recall. Without these features, the system would be neither viable nor functional. This encompasses user identification, resource management, flashcard production, and rudimentary review functionality

The should have criteria improve usability and system administration, although they are not necessary for a minimum viable system. For example, contextual navigation from flashcards to specific sections of a resource improves learning effectiveness but may be limited by technical restrictions such as document format or annotation capability.

The could have requirements offer value but are considered lower priority within the project scope. Revision tracking can facilitate reflection and long-term participation but is not required for demonstrating fundamental system functionality.

#### 4.2.4 Non-functional Requirements

| ID    | Non-Functional Requirement | Description                                                                                                                                                                                                | Prioritisation |
|-------|----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| NFR1  | Usability                  | The system must be intuitive and easy to use for students with minimal technical knowledge, allowing them to request for resources to be uploaded, create flashcards, and review content without training. | Must have      |
| NFR2  | Usability                  | Navigation between the research repository and flashcards should be clear, consistent, and logically structured.                                                                                           | Must have      |
| NFR3  | Performance                | Core system actions should complete within a few seconds under normal usage conditions.                                                                                                                    | Must have      |
| NFR4  | Security                   | User passwords must be stored securely using hashing, and access control must distinguish between standard users and administrators.                                                                       | Must have      |
| NFR5  | Privacy                    | The system should comply with UK data protection principles and GDPR, collecting only necessary user data.                                                                                                 | Must have      |
| NFR6  | Reliability                | User data must be saved correctly and remain accessible across sessions.                                                                                                                                   | Must have      |
| NFR7  | Error Handling             | The system should provide clear and informative error messages when operations fail.                                                                                                                       | Should have    |
| NFR8  | Accessibility              | The interface should use readable fonts, clear contrast, and a simple layout to support accessibility and inclusiveness.                                                                                   | Should have    |
| NFR9  | Maintainability            | The system should be implemented using modular and well-structured code to allow future modification and extension.                                                                                        | Should have    |
| NFR10 | Scalability                | The system should allow future features to be added without major redesign.                                                                                                                                | Could have     |

##### 4.2.4.1 Non-functional Requirements Justification

The non-functional requirements were prioritised to guarantee that the system provides a high-quality user experience while ensuring security, dependability and long-term viability. Usability (NFR1, NFR2) was classified as a ‘Must Have’ requirement because the system is intended for students with varying levels of technical expertise, and an unintuitive interface would undermine adoption and



effectiveness. Performance (NFR3) was also prioritised as ‘Must Have’ as delays in core actions such as searching resources or reviewing flashcards could negatively impact study efficiency. Security and privacy (NFR4, NFR5) were identified as ‘Must Have’ due to the storage of user credentials and academic materials, requiring secure password hashing, role-based access control, and compliance with UK data protection principles. Reliability (NFR6) was prioritised to prevent data loss and ensure that user content remains consistently accessible across sessions. Additional requirements such as error handling, accessibility and maintenance (NFR7-NFR9) were categorised as ‘Should Have’ because they enhance usability, inclusiveness and long-term code quality, although the system remains functional without full optimisation in these areas. Scalability (NFR10) was designated as ‘Could Have’ to allow for future feature expansion without requiring architectural redesign. Collectively, these non-functional requirements ensure that the system is not only functional but also secure, usable and sustainable.

#### **4.2.4.2 Prioritisation Rationale**

The MOSCOW technique was used to prioritise non-functional requirements to reflect the projects core objectives and limitations. Must have requirements that focus on usability, performance, security, privacy and reliability, as these qualities are required for the system to function as a usable academic tool. If these requirements are not met, students may find it difficult to use the system, they could encounter delays during revision activities, or have a risk of losing personal study data, undermining the systems’ main objective of promoting effective information organisation and active recall. Should have requirements enhance the overall user experience and long-term sustainability of the system but are not required for a minimum viable prototype. Improved error handling, accessibility considerations and maintainable code structure, contributes to usability, inclusivity, and ease of future development while remaining flexible within the projects restricted time and technical scope. Could have requirements reflect desirable extensions, such as scalability for future developments, which are acknowledged as valuable but intentionally deprioritised to guarantee that core functionality is delivered successfully within the project’s timeframe.

### **4.3 Design**

The integrated research repository and flashcard system was designed using an iterative and evidence-informed process based on findings from the literature research, requirements analysis, and product comparison. Rather than moving directly into implementation, the design approach progressed through the conceptual modelling, functional modelling, data modelling and interface prototyping. This layered approach meant that both the broader problem environment and the systems technical structure were thoroughly understood prior to development.

All design decisions were based on a user-centered approach. The system was not created just as a database application or revision tool, but rather as an integrated academic workflow support system. Early conceptual investigation with a rich picture helped to identify stakeholders' needs and systemic challenges. Formal UML use case modelling then converted these requirements into functional behaviours. The Entity-Relationship Diagram defined the data architecture needed to enable these behaviours, and Axure interface prototypes validated usability and navigational clarity before coding. This structured progression from abstract to technical design meant that the literature findings, requirements and implementation decisions were all traceable.

#### 4.3.1 Conceptual Design Rich Picture

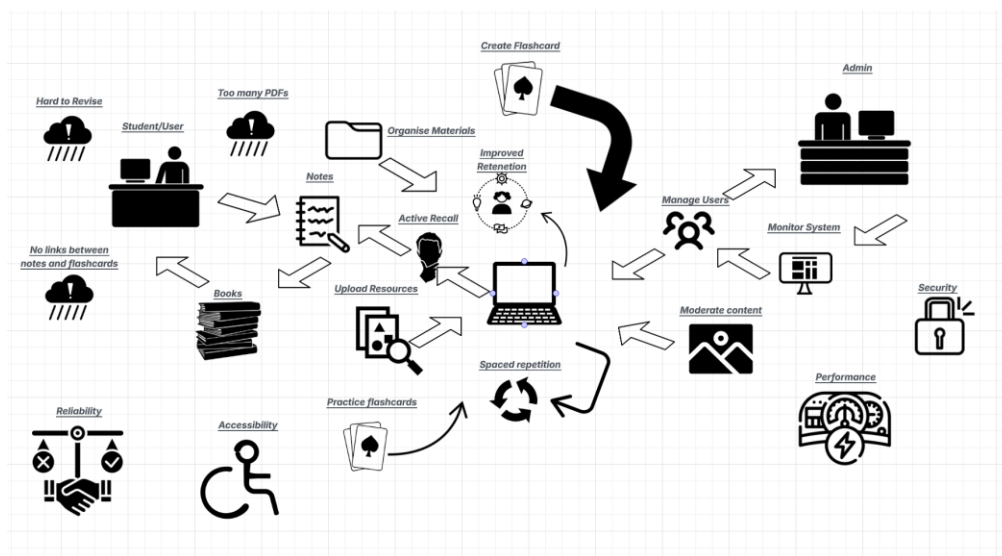


Figure 6

The rich picture (Figure 6) was utilised to investigate the problem space in a comprehensive and non-technical manner. Unlike formal modelling tools, the rich picture includes contextual aspects including student frustration, environmental constraints and fragmented digital practices. This was especially useful since the research recognised information overload and tool fragmentation as systemic rather than just technological concerns.

The diagram illustrates students interacting with a variety of disconnected tools and highlights key issues such as difficulty locating materials during revision, loss of contextual understanding, and inefficient switching between systems. It also denotes administrative control, data security concerns, and performance expectations.

Visualising these relationships revealed that the main issue was fragmentation rather than a lack of tools. This directly influenced the design decisions to combine repository and flashcard functionality into a single system rather than having separate apps. The rich picture therefore provided as a link between theoretical discoveries and real system architecture.

#### 4.3.2 Functional Design – UML Use Case Model

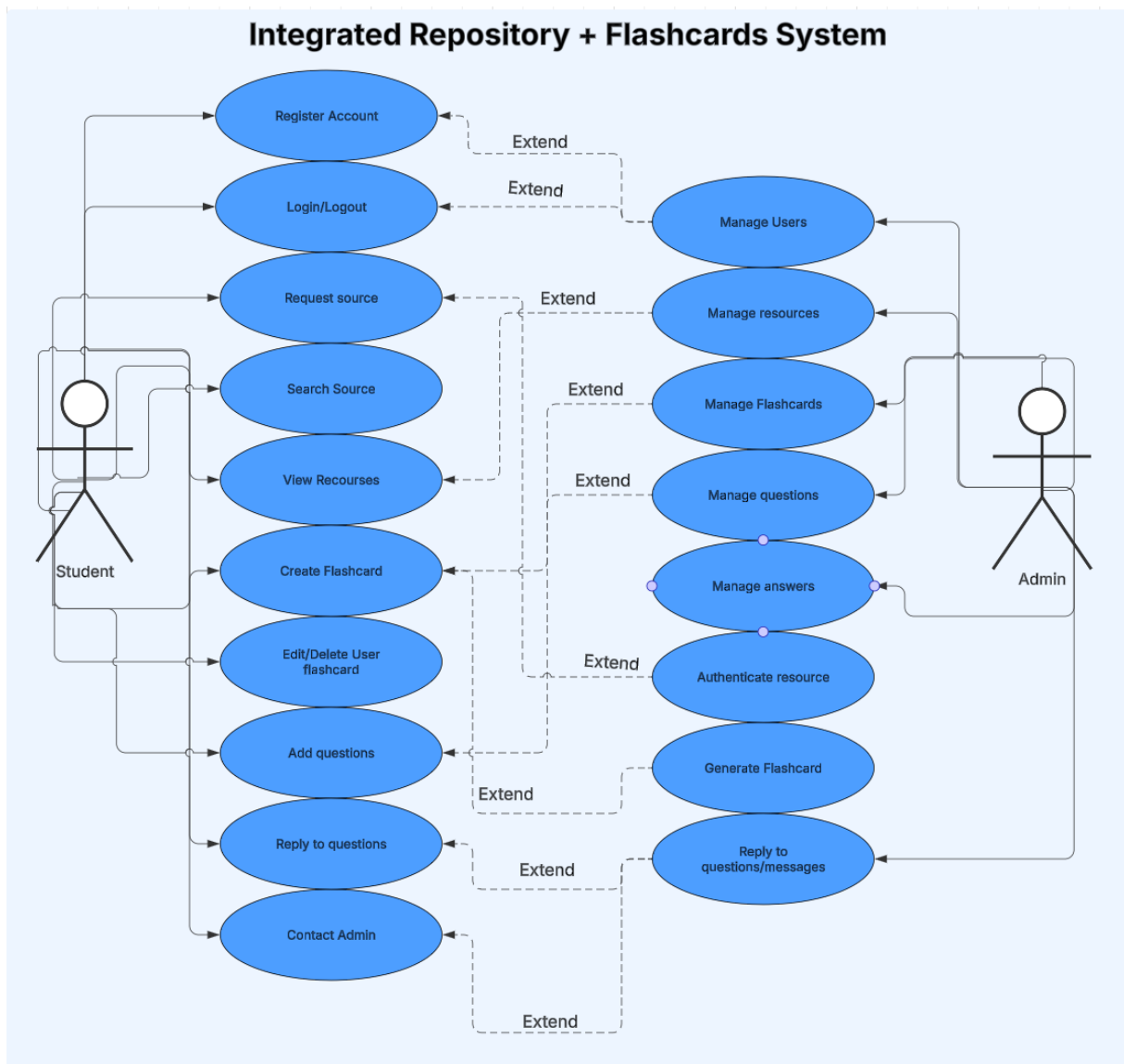


Figure 7

The UML use case diagram (Figure 7) formalises the systems functional scope while also providing a structured depiction of user interactions. Two key factors identified, the student/user and the administrator. This separation reflects the role-based access control concepts and is consistent with LSEPi objectives for accountability and data protection.

Prioritised functional needs were identified into use cases such as viewing recourses, making flashcards and evaluating information. The inclusion of contextual navigation as a separate use case indicates the projects emphasis on preserving conceptual relationships between revision materials and academic resources.

UML notation was chosen because it is extensively used in software engineering practice and promotes clarity, consistency and traceability. The use case model guarantees that each developed feature can be linked to a specific need, avoiding ambiguity and allowing for systematic testing later in the project.

### 4.3.3 Data Design – Entity Relationship Diagram

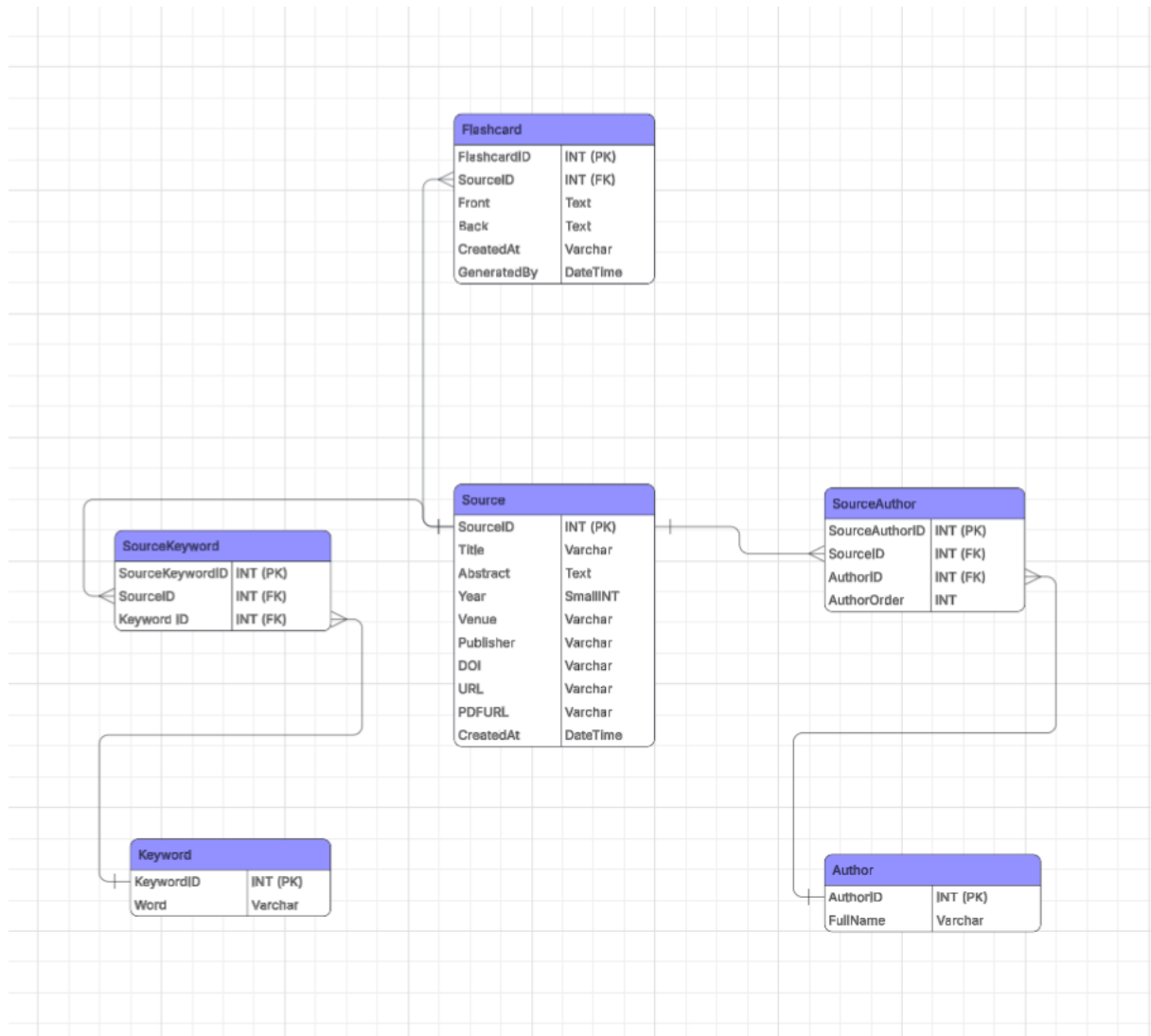


Figure 8

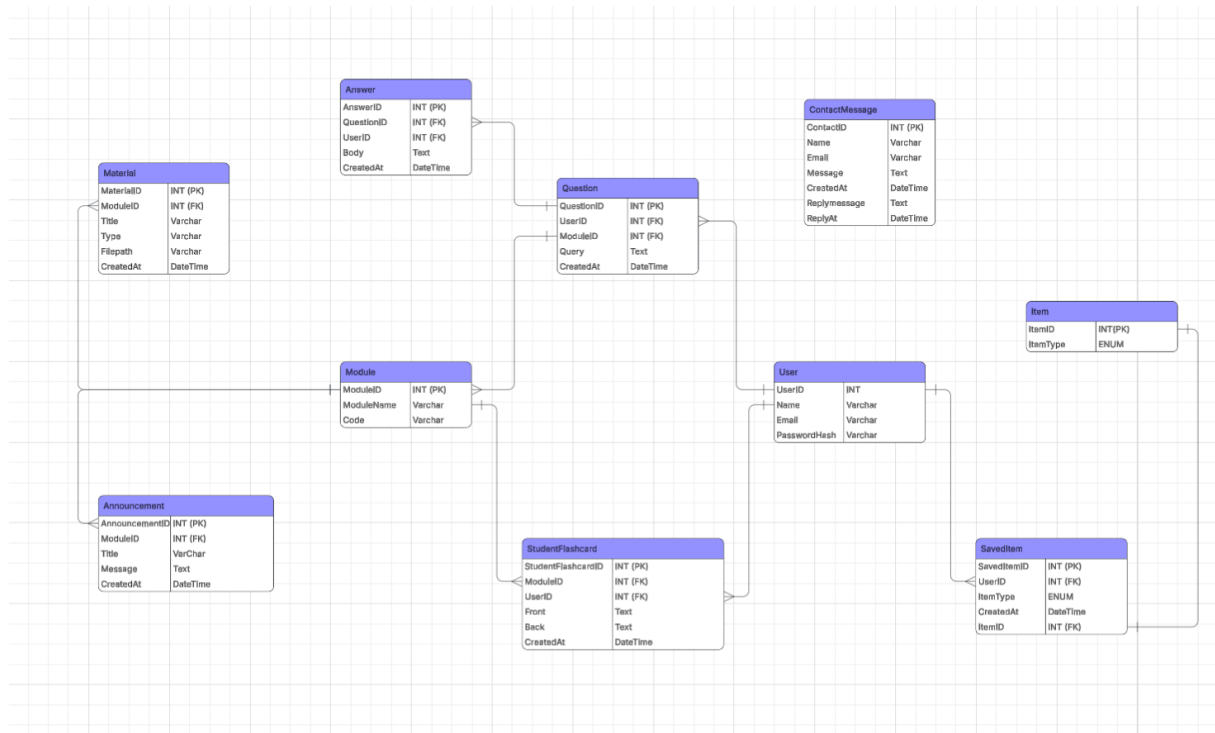


Figure 9

The ERD (Figure 8 & 9) specifies the logical data architecture necessary to support system functionality. The core entities are user, source, author, keyword, flashcard, module, question, answer and flashcard. For improvement of the system functionality and user friendliness and satisfaction the following entities were created material, announcement, saved item, item, student flashcard and contact message. Relationships were represented via crows foot notation to express cardinality and participation limitations.

Many-to-many links between source and author, as well as source and keyword, were handled using junction tables. This method assures database normalisation and avoids duplication of author and keyword information. Composite primary keys were employed in junction tables guarantee referential integrity and prevent redundancy.

The data model was created to comply with the Third Normal Form, which ensures that non-key characteristics rely entirely on the primary key. This lowers update anomalies while improving scalability. The use of contextual linking between flashcard and source entities guarantees that revision information remains directly associated with academic material, promoting deeper learning rather than separate memorisation.

The ERD was created directly from functional requirements, ensuring that data structure and user interaction are consistent.

#### 4.3.4 Navigational Structure and Interface Design



Figure 10

The navigational layout was created to reduce cognitive load while providing straightforward access to important functionality. The Axure homepage prototype (Figure 10) shows a centralised structure with easily identifiable navigation buttons for sources, modules, flashcards, questions and latest uploads. (See Appendix b1.1 – b2.7 for more page mid-fidelities and a low fidelity for the navigation layout) The top-level horizontal navigation layout was chosen over the sidebar or multi-level dropdown options. This choice was taken to remove hidden navigation layers while maintaining visibility of essential system functionalities. The search bar's primary placement emphasises the necessity of quick information retrieval during revision. Visual design concepts including simplicity, consistency, spacing and clear typography were prioritised. The interface minimises excessive graphical complexity, which is consistent with research showing that overly cluttered digital environments contribute to cognitive overload. Alternative designs, such as dashboard widget layouts and collapsible side menus, were considered but discarded due to increased interface complexity and reduced clarity for first-time users.



Figure 11

The admin section's navigational interface (Figure 11) was designed to be simple, allowing the administrator to quickly access all major features and capabilities necessary to properly manage the system. The UI has clearly marked navigation buttons for sources, modules, flashcards, questions, and newest uploads, allowing the administrator to swiftly switch between regions without unnecessary complication. This design decreases cognitive strain and promotes effective job completion, especially for administrative tasks like managing users, evaluating material, and maintaining system data.

The decision to employ a simple, centralised style was informed by usability concepts found in the literature, which promote clarity, consistency, and simplicity of navigation. By eliminating multi-layered menus and hidden navigation structures, the design always keeps all key functions visible and accessible. This is especially crucial for administrative users who need to handle the system quickly and consistently. Overall, the structure prioritises efficiency and usability, allowing the administrator to do tasks with minimum effort while yet having a comprehensive overview of the system's fundamental components.

#### 4.3.5 User Interface Design Nielsen Heuristics

Nielsen's usability heuristics are a generally regarded set of principles for evaluating and improving user interface design (Nielsen, 1994). In this project, these heuristics are used to guide the creation of an intuitive and user-friendly system. For example, clear feedback messages when user's complete activities such as signing in or generating flashcards help improve system visibility. A similar design and navigation structure across all pages ensures consistency and standards, allowing users to predict system performance with ease. Recognition rather than recall is achieved by giving visible navigation

choices and search capabilities which reduce the need for users to remember where information is kept.

#### **4.3.6 Creative Methods and Alternatives Considered**

Before finalising on the integrated architecture, several other design methods were considered. One alternative was to keep separate repositories and flashcard systems connected via API integration. While technically viable, this approach was rejected because it would exacerbate the fragmentation identified in the research and require users to switch context more frequently.

Another alternative was to create automated flashcards utilising AI-driven concept extraction. While creative, this approach was deemed beyond the scope of the minimum viable prototype and risked inaccuracies in automatically generated material. Manual flashcard production with contextual linking was preferred for dependability and user control. As future development looking into using AI-driven flashcard generation, as it would allow the flashcard to have more in depth knowledge on each source. The final design prioritises integration, simplicity, and traceability over technological innovation.

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#### **4.3.7 LEPSi Considerations in Design**

Legal, social, ethical and professional issues were included into the design process. Secure authentication and password hashing were used to secure user data. Role-based access control guarantees that administrative privileges are properly controlled.

Data protection considerations influenced decisions for storing limited personal data. Only basic user information such as email, addresses and login passwords is stored. Accessibility considerations, such as readable fonts and a logical layout structure, supports inclusive design.

These decisions demonstrate professional responsibility in system development and are consistent with legislative obligations under UK data protection legislation.

### **4.4 Conclusions**

In conclusion, this chapter has defined and structured functional and non-functional requirements that are required to achieve the project's aim of combining a research repository and flashcard-based learning/revision. The analysis highlighted the significance of integrating effective information management with active recall techniques, leading to core requirements such as resources organisation, flashcard creation, contextual linking and user authentication. These criteria significantly influenced design decisions, such as the relational database structure, UML use case modelling and a



streamlined user-centered interface. The use of an integrated system design was a deliberate response to the usability gap identified in both the literature review and product comparisons, in which existing technologies fail to provide a unified workflow.

The design approach contributes to the overall project objectives by ensuring usability, performance, security and scalability are embedded into the system from the outset. While certain limitations were recognised, such as scope constraints that prevented advanced features like AI generated flashcards or large-scale user validation, these were acknowledged and addressed by prioritisation utilising the MoSCoW technique and consideration for future development.

Overall, the structured and iterative analysis and design process ensures that the proposed system is both technically viable and meets user requirements. The following chapter will expand on this by discussing implementation, explaining how the defined requirements and design decisions are translated into a functioning application.

## 5 IMPLEMENTATION

This chapter describes the proposed system's implementation, including how the preceding chapter's design and requirements were converted into a functioning prototype. It describes the entire system architecture, development process, and technologies utilised to create the application, including the database and user interface components. The chapter begins by explaining the codebase's structure and the organisation of system files, then moves on to examine the three-tier architecture used to separate display, application logic, and data management. Key features such as resource management, flashcard creation, contextual navigation, and user authentication are then demonstrated to show how the system supports both academic organisation and active recall. Finally, the chapter discusses the testing methodologies used to ensure system stability and accuracy, demonstrating that the implementation fits the functional and non-functional criteria stated previously.

### 5.1 Code Repository Outline

The system was developed entirely from scratch using PHP, Html and MySQL, with a modular and organised approach to increase maintainability and clarity. The project is divided into multiple directories to separate concerns between logic, presentation and style.

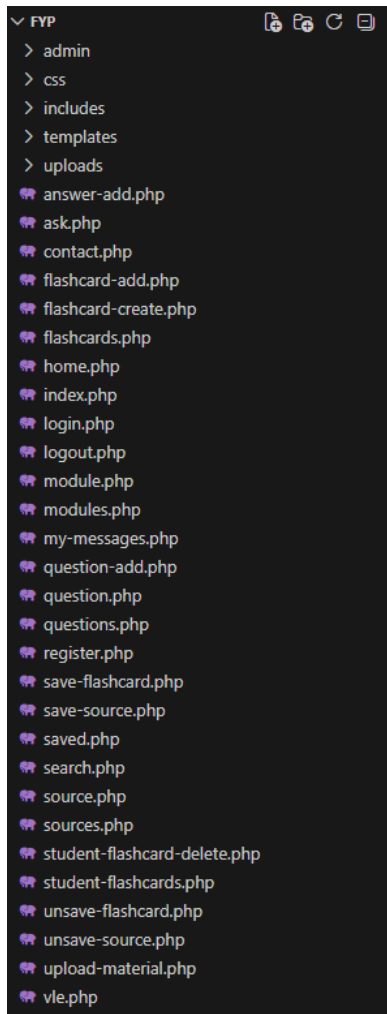


Figure 12

The main structure of the system (Figure 12) includes the applications main entry point as `home.php`, while `includes` contains reusable components like `DatabaseConnection.php` and helper functions. `Templates` contains Html templates for dynamic pages, and `CSS` contains stylesheets for visual design and layout. Feature specific files include `sources.php` for displaying and managing academic sources, `flashcard.php` for flashcard creation and review and `questions.php` for the forum/support. There are a lot more feature specific files for different functionalities like `save-flashcard.php` which allows flashcards to be saved for revision. The system accesses the database via PDO, which allows prepared statements to be used for all queries. This enables secure handling of user inputs and lowers the risk of SQL injection issues. The core implementation did not include any extra libraries or frameworks, ensuring that the system reflects the original development work.

## 5.2 System Architecture and Implementation Approach

The system was developed with a three-tier architecture consisting of the presentation layer, application logic layer and database layer. This architectural style was chosen to provide a clear separation of responsibilities, improving maintainability and ease of debugging.

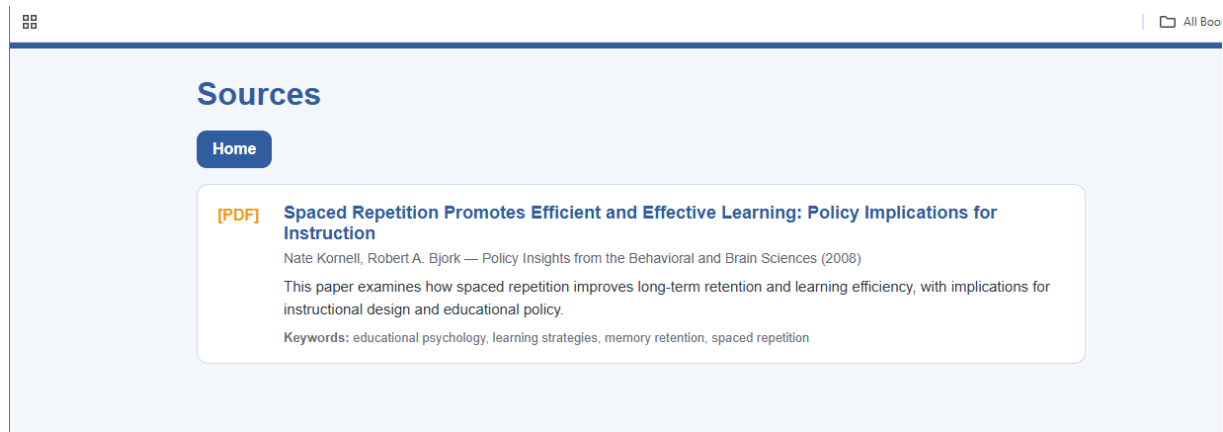


Figure 13

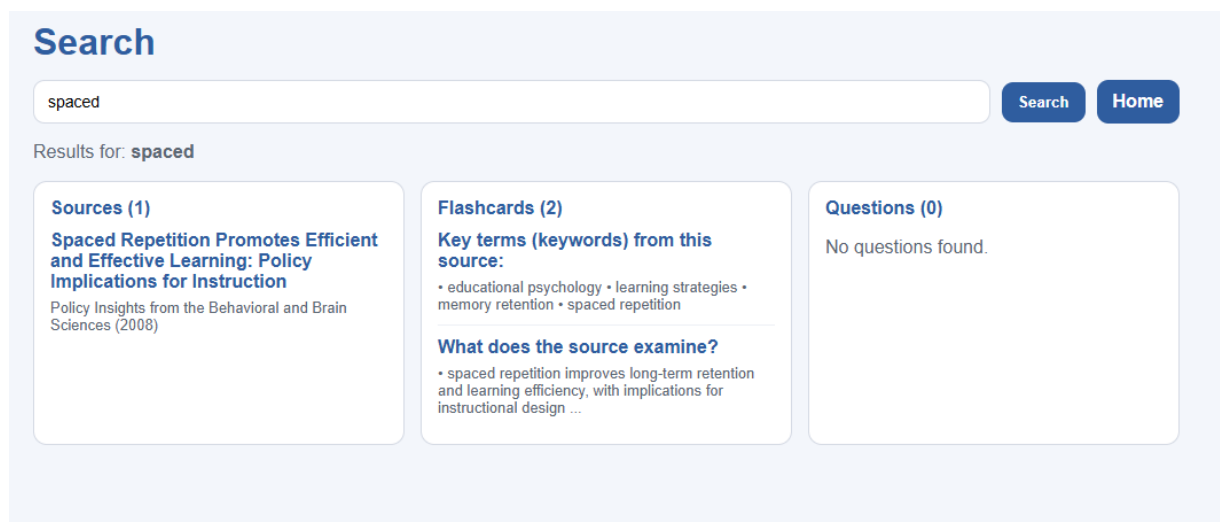


Figure 14

The presentation layer handles user interaction and interface rendering. It was implemented with HTML and CSS and PHP was used to dynamically generate content based on user input. This is illustrated in Figures 13 & 14, where academic resources and search results are displayed dynamically. These interfaces demonstrate how data retrieved from the database is presented to the user in a structured and accessible format. The application logic layer handles user requests and manages system behaviour. This includes handling form inputs, verifying user input, running database queries, and maintaining session-based authentication. When a user performs a search (Figure 14), the input is processed by the application layer, which queries the database and returns relevant results to be displayed. This demonstrates the role of the application layer in coordinating communication between the interface and the database. The implementation of the search functionality further demonstrates how the application layer processes user input and coordinates database interactions. When a user enters a keyword, the system validates it and converts it into a wildcard format for partial matching. It then runs queries against several tables, including sources, flashcards, and questions. An extract of the implementation is shown below.

```

20 if ($q != '') {
21     $like = '%' . $q . '%';
22
23
24     if (tableExists($pdo, 'Source')) {
25         $sql = "
26             SELECT SourceID, Title, Year, Venue
27             FROM Source
28             WHERE Title LIKE :q OR Abstract LIKE :q OR DOI LIKE :q
29             ORDER BY SourceID DESC
30             LIMIT 20;
31         ";
32         $stmt = $pdo->prepare($sql);
33         $stmt->execute([':q' => $like]);
34         $results['sources'] = $stmt->fetchAll(PDO::FETCH_ASSOC);
35     }
36
37
38     if (tableExists($pdo, 'Flashcard')) {
39         $sql = "
40             SELECT FlashcardID, Front, Back
41             FROM Flashcard
42             WHERE Front LIKE :q OR Back LIKE :q
43             ORDER BY FlashcardID DESC
44             LIMIT 20;
45         ";
46         try {
47             $stmt = $pdo->prepare($sql);
48             $stmt->execute([':q' => $like]);
49             $results['flashcards'] = $stmt->fetchAll(PDO::FETCH_ASSOC);
50         } catch (PDOException $e) {
51             $results['flashcards'] = [];
52         }
53     }
54 }
55
56
57 if (tableExists($pdo, 'Question')) {
58     $sql = "
59         SELECT QuestionID, Query, CreatedAt
60         FROM Question
61         WHERE Query LIKE :q
62         ORDER BY QuestionID DESC
63         LIMIT 20;
64     ";
65     $stmt = $pdo->prepare($sql);
66     $stmt->execute([':q' => $like]);
67     $results['questions'] = $stmt->fetchAll(PDO::FETCH_ASSOC);
68 }
69 }

```

Figure 15

This implementation (Figure 15) uses parameterised queries to securely handle user input and prevent SQL injection. The use of wildcard matching enables flexible keyword-based searching, while limiting the number of results improves performance. Similar reasoning is applied to flashcards and questions, allowing a unified search across the system. This shows how the application layer handles user requests and communicates with the database layer to produce dynamic output for the presentation layer. The database layer uses MySQL to manage persistent data storage. It holds structured data about users, academic materials, flashcards and their relationships. The database architecture adheres to the ERD created in the previous chapter, ensuring efficient data retrieval and integrity. The results displayed in figures 13 and 14 confirm that data is successfully retrieved and integrated into the user interface. This architecture allows for seamless interaction between system components. For example, when a user searches for a resource, the application layer processes the request, and retrieves data from the database, and displays the results through the presentation layer.

## 5.3 Database Implementation

The database was implemented using MySQL, based directly on the Entity Relationship Diagram developed during the design phase. The structure follows normalisation principles to reduce redundancy and maintain data integrity. The database design is structured around several key entities that support the systems fundamental functionality. The User table holds user credentials and role information, allowing for authentication and access control. The Source table contains academic materials, including title, abstract and associated links, and serves as the repository's foundation. The Flashcard table contains question and answer pairs that are explicitly connected to sources, promoting active recall while keeping contextual relevance. The Module table is used to organise academic information, allowing users to properly manage their materials. In addition, the Question-and-Answer tables enable users to engage and exchange information inside the system.

| #                        | Name             | Type    | Collation | Attributes | Null | Default | Comments | Extra          | Action             |
|--------------------------|------------------|---------|-----------|------------|------|---------|----------|----------------|--------------------|
| <input type="checkbox"/> | 1 SourceAuthorID | int(11) |           | UNSIGNED   | No   | None    |          | AUTO_INCREMENT | Change  Drop  More |
| <input type="checkbox"/> | 2 SourceID       | int(11) |           |            | No   | None    |          |                | Change  Drop  More |
| <input type="checkbox"/> | 3 AuthorID       | int(11) |           |            | No   | None    |          |                | Change  Drop  More |
| <input type="checkbox"/> | 4 AuthorOrder    | int(11) |           |            | No   | None    |          |                | Change  Drop  More |

Figure 16

To support these fundamental entities, various additional tables are used to handle relationships and improve functionality. The SourceAuthor table (Figure 16) resolves the many-to-many relationship between sources and authors, ensuring proper attribution avoiding data duplication. Similarly, the SourceKeyword table links sources to keywords, allowing for more effective content searching and organisation. The SavedItem table enables users to bookmark important materials or flashcards, increasing accessibility and supporting personalised study workflows.

| Foreign key constraints |                                                                |          |                                                   |
|-------------------------|----------------------------------------------------------------|----------|---------------------------------------------------|
| Actions                 | Constraint properties                                          | Column   | Foreign key constraint (INNODB)                   |
|                         |                                                                |          | Database Table Column                             |
| Drop                    | fk_sourceauthor_author<br>ON DELETE CASCADE ON UPDATE RESTRICT | AuthorID | sources author AuthorID                           |
| Drop                    | fk_sourceauthor_source<br>ON DELETE CASCADE ON UPDATE RESTRICT | SourceID | sources source SourceID                           |
| Drop                    | sourceauthor_ibfk_1<br>ON DELETE CASCADE ON UPDATE RESTRICT    | SourceID | sources source SourceID                           |
| Drop                    | sourceauthor_ibfk_2<br>ON DELETE CASCADE ON UPDATE RESTRICT    | AuthorID | sources author AuthorID                           |
|                         | Constraint name<br>ON DELETE RESTRICT ON UPDATE RESTRICT       |          | sources <input type="text"/> <input type="text"/> |
| + Add constraint        |                                                                |          |                                                   |

Figure 17

Foreign key constraints (Figure 17) were used extensively to maintain referential integrity between tables. This ensures that each author is correctly associated with a valid source. During development, several challenges were encountered related to foreign key constraints and data consistency. For

instance, errors occurred when attempting to insert data that did not yet exist in referenced tables. These issues were resolved by refining the database structure and ensuring correct insertion order, demonstrating the importance of careful database design.

## 5.4 Core System Features

### 5.4.1 Resource Management

The resource management feature (Figure 13) enables users to organise, search and read academic content from a single repository. Each resource contains metadata such as title, abstract, authors and links to other resources or PDF documents. This organised method allows users to store and manage enormous amounts of academic knowledge in an organised and accessible manner. Resources may be classified using modules or keywords, allowing users to group similar resources and increase retrieval performance. Implementing a keyword-based search engine (Figure 14) allows users to easily find relevant materials by searching titles, abstracts or related identifiers. This capability immediately addresses the issue of information overload that was raised in previous chapters by minimising the time and effort necessary to access study resources.

### 5.4.2 Flashcard System

The screenshot displays a web interface for a flashcard system. At the top, there is a 'Back to Source' link and a 'Generate Flashcards' button. Below this, the source text is: 'Spaced Repetition Promotes Efficient and Effective Learning: Policy Implications for Instruction'. The interface shows four flashcards, each with a question (Q), an answer (A) containing bulleted keywords, a 'Saved' status with a green checkmark, and a generation timestamp 'Generated: RULES • 2026-04-07 17:57:33'.

← Back to Source

**Flashcards** Generate Flashcards

Source: Spaced Repetition Promotes Efficient and Effective Learning: Policy Implications for Instruction

**Q:** Key terms (keywords) from this source:

**A:** • educational psychology  
• learning strategies  
• memory retention  
• spaced repetition

Saved ✓ Generated: RULES • 2026-04-07 17:57:33

**Q:** What implications are mentioned?

**A:** • instructional design  
• educational policy

Saved ✓ Generated: RULES • 2026-04-07 17:57:33

**Q:** What benefits or findings are mentioned?

**A:** • improves long-term retention and learning efficiency, with implications for instructional design and educational policy  
• improves long-term retention  
• improves learning efficiency  
• improves with implications for instructional design

Saved ✓ Generated: RULES • 2026-04-07 17:57:33

**Q:** What does the source examine?

**A:** • spaced repetition improves long-term retention and learning efficiency, with implications for instructional design and educational policy

Saved ✓ Generated: RULES • 2026-04-07 17:57:33

Figure 18

← Back to hub

## Create Flashcards

Creating as Richard

**Module**

Select a module... ▼

**Question**

**Answer**

**Save Flashcard**

Figure 19

The flashcard system (Figure 18 & 19) allows users to create and manage revision materials for active recall. Users can generate flashcards manually or base them on the stored academic sources, ensuring that revision content is directly linked to relevant study material. Each flashcard contains a question and answer, allowing users to assess their understanding of key concepts. Flashcards are saved in a database and linked to specific sources ensuring that revision is contextually relevant rather than isolated. This linking improves learning efficacy by allowing users to return to the original content as needed.



### 5.4.3 Contextual Navigation

#### Saved Flashcards (4)

##### What does the source examine?

- spaced repetition improves long-term retention and learning efficiency, with implications for instructional design ...

From: [Spaced Repetition Promotes Efficient and Effective Learning: Policy Implications for Instruction](#)

##### What benefits or findings are mentioned?

- improves long-term retention and learning efficiency, with implications for instructional design and educational po...

From: [Spaced Repetition Promotes Efficient and Effective Learning: Policy Implications for Instruction](#)

##### What implications are mentioned?

- instructional design • educational policy

From: [Spaced Repetition Promotes Efficient and Effective Learning: Policy Implications for Instruction](#)

##### Key terms (keywords) from this source:

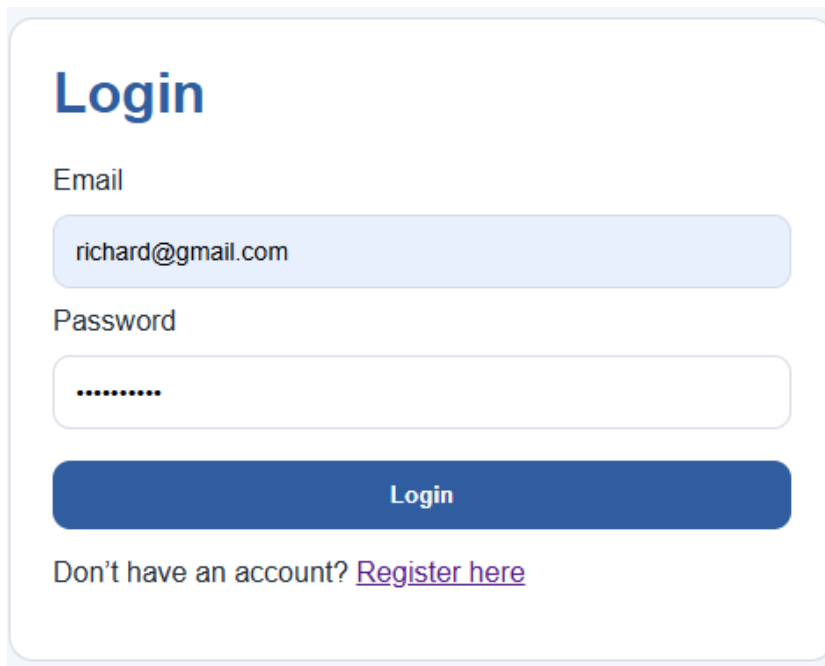
- educational psychology • learning strategies • memory retention • spaced repetition

From: [Spaced Repetition Promotes Efficient and Effective Learning: Policy Implications for Instruction](#)

Figure 10

Contextual navigation (Figure 20) is a fundamental aspect of the system, allowing users to move directly from a flashcard to its academic source. This feature guarantees that revision is not isolated from the original learning materials, solving a significant problem seen in prior flashcard platforms. This feature demonstrates the integration of several system components and emphasises the benefits of merging repository and flashcards functionality within a single platform. It reduces the need for users to transition between several tools, increasing workflow efficiency and lowering cognitive load.

#### 5.4.4 User Authentication and Roles



**Login**

Email

richard@gmail.com

Password

.....

Login

Don't have an account? [Register here](#)

Figure 11

```
login.php
1 <?php
2 session_start();
3 require_once 'includes/DatabaseConnection.php';
4
5 $error = '';
6
7 if ($_SERVER['REQUEST_METHOD'] === 'POST') {
8     $email = trim($_POST['Email'] ?? '');
9     $password = $_POST['Password'] ?? '';
10
11     if ($email === '' || $password === '') {
12         $error = 'Please enter both email and password.';
13     } else {
14         $stmt = $pdo->prepare("SELECT UserID, Name, Email, PasswordHash FROM user WHERE Email = :email LIMIT 1");
15         $stmt->execute([':email' => $email]);
16         $user = $stmt->fetch(PDO::FETCH_ASSOC);
17
18         if ($user && password_verify($password, $user['PasswordHash'])) {
19             $_SESSION['user_id'] = $user['UserID'];
20             $_SESSION['user_name'] = $user['Name'];
21             $_SESSION['user_email'] = $user['Email'];
22
23             if ($user['Email'] === 'admin@gmail.com') {
24                 header('Location: admin/home.php');
25                 exit;
26             }
27
28             header('Location: home.php');
29             exit;
30         } else {
31             $error = 'Invalid email or password.';
32         }
33     }
34 }
35
36 $title = 'Login';
37 $stylesheets = ['css/login.css'];
38
39 ob_start();
40 include 'templates/login.html.php';
41 $output = ob_get_clean();
42
43 include 'templates/layout.html.php';
```

Figure 12

The system uses user authentication to guarantee safe access and a personalised user experience. When a user enters their login information (Figure 21 & 22), the program validates the input and queries the database to obtain the appropriate user record based on the given email address. The system then compares the entered password to the stored hashed password using a safe verification process. If the

credentials are correct, session variables are established to save important user information such as the user ID, name and email address, allowing the system to maintain an authorised session across several pages. Role-based access control works by differentiating between normal users and administrators. This is achieved with conditional logic that validates the user's email and leads administrators to a separate admin page, while normal users are forwarded to the main system interface. This separation guarantees that administrative functionality, such as user and content management, is only available to authorised users.

## **5.5 Testing**

Testing was conducted to verify the system meets both the functional and non-functional requirements, ensuring that the prototype works properly, reliably and is usable by its intended users. A structured testing strategy was used, which included white box and black box testing, integration testing and usability testing. This multi-layered approach ensured that both the systems underlying logic and the external user experience were thoroughly tested. The testing process was guided by a test plan based on the functional and non-functional requirements identified earlier in the project. The system functionalities were tested including resource management, flashcard generation, contextual navigation and user authentication. Testing focused on crucial system components rather than all minor aspects to guarantee thorough validation. Appendix C and D contain extensive test cases and full findings, however this part provides a concise summary of the testing methods, justification and key outcomes.

### **5.5.1 Unit Testing (White Box Testing)**

Unit testing was conducted using a white box approach to assess the core functionality of individual system components. This involved testing backend procedures including database queries, input validation, and session management. The testing aimed to ensure that essential activities such as database insertion and retrieval for sources and flashcards, enforcement of foreign key relationships, management of duplicate data entries, and session-based authentication logic functioned correctly. For example, tests were performed to ensure that when a user generates a flashcard, the system successfully puts data into the database while preserving the link with the relevant source. Similarly, input validation was conducted to ensure that invalid data did not corrupt system records. During development several issues were identified through unit testing including foreign key constraint errors and data type incompatibilities. These issues were resolved by refining the database schema and improving query logic. As a result, unit testing guaranteed that individual components worked properly and consistently before they were integrated into the larger system.

### **5.5.2 Black Box Testing**

Black Box Testing was performed to assess the system from the user's perspective, without regard to internal implementation. This approach focused on ensuring that the system produces the proper outputs for the given inputs. Test cases were created based on key functional requirements such as user registration and login, resource upload and display, search functionality, flashcard generation and review, and navigation between flashcards and their sources, it is properly stored and shown in the repository. Similarly, the search functionality was tested to verify that appropriate results are provided based on keyword input and flashcard creation was confirmed to ensure that flashcards are connected and displayed properly. Screenshots were captured to show successful outcomes, such as proper search results and navigation flows and to provide visual evidence that the system performs as intended from a user perspective. The findings showed that the majority of fundamental functionality work properly, while minor concerns such as edge case input handling were identified and resolved during development.

### **5.5.3 Integration Testing**

Integration testing was conducted to guarantee that the systems different components function together properly. This testing focused on validating interactions between the presentation layer, application logic layer and database layer. The process included ensuring that uploaded resources are correctly stored in the database and displayed on the user interface, that flashcards created from resources are accurately retrieved and retain their links to the original sources, and that navigation between pages is seamless. This includes transitions like switching from a source to its corresponding flashcards and vice versa. The results of integration testing showed that the system works as a coherent application rather than a collection of separate components. Furthermore, this testing validated the efficacy of the three-tier architecture by demonstrating that communication between system layers works perfectly and supports the expected system functionality.

### **5.5.4 Usability Testing**

Usability testing was carried out to see how readily users could engage with the system. Testing was carried to perform key tasks such as searching a source, creating flashcards and locating revision materials. Feedback indicated that the system was generally straightforward and simple to use, thanks to its clear layout and centralised navigation structure. Users were able to accomplish tasks with little assistance, indicating that the system satisfies usability standards. Some minor improvements were discovered, such as increasing visual indicators for flashcard-source linking and clarifying specific labels. These were included into the system design to enhance the user experience. Screenshots in Appendix D are provided to demonstrate essential interactions and show that the system offers efficient and user-friendly workflows.

## 5.6 Conclusion

Finally, the testing procedure shows that the system satisfies most of its functional and non-functional requirements. The combination of unit testing, black box testing, integration testing and usability testing resulted in a thorough evaluation of system performance and user experience. The findings confirm that fundamental capabilities such as resource management, flashcard generation and contextual navigation work as expected and align with the project's goals. Testing also revealed minor flaws, such as managing edge situations and the lack of advanced functionality, which are acknowledged as topics for future development. Overall, the testing approaches were effective at validating the system and identifying opportunities for improvement. The prototype offers a dependable and useable solution to the challenge outlined in previous chapters. The following chapter will on these results by conducting a critical examination of the systems overall efficacy.

## **6 RESULTS AND EVALUATION**

### **6.1 Introduction**

This chapter presents the project's outcomes and conducts a critical evaluation of the final prototype in respect to the original intent, objectives, and requirements specified in earlier chapters. The goal of this chapter is to illustrate what was accomplished throughout the development process, evaluate how successfully the system tackles the problem of fragmented study tools and information overload, and reflect on the project's overall success and limits.

### **6.2 Results**

This section presents the results of the produced prototype in a structured and objective manner. The findings are closely related to the project's goal, which was to create an integrated research library and flashcard system to enhance organisation, decrease information overload, and promote active recall.

#### **6.2.1 Implementation of Core System Features**

The final prototype effectively meets the primary functional requirements identified during the requirements analysis stage. These include user authentication, resource management, search functionality, flashcard creation, contextual navigation, and an administration interface.

The resource management system enables the effective storage, organisation, and retrieval of academic materials. Each resource has structured metadata such as the title, abstract, authors and external links. This significantly improves on previous repository systems by allowing organised storage within a single system rather than across numerous platforms. The incorporation of keyword-based search functionality enables users to quickly obtain, relevant materials, addressing the literature identified issue of information overload. The flashcard system was also successfully developed, allowing users to create question and answer pairings that are directly connected academic sources. This ensures that revision materials are not isolated but remain contextually linked to the original learning content. This is an important research achievement since it immediately addresses then identified gap in existing tools, which allows revision systems and repositories to operate independently. The introduction of contextual navigation is a particularly significant effect. Users may navigate straight from a flashcard to its corresponding academic source, allowing them to examine the material's original context. This feature demonstrates the integration of system components and promotes deeper learning, as highlighted in retrieval practice literature. User authentication and role-based access control were both effectively implemented. The system provides secure login functionally using hashed passwords and session-based authentication, as well as differentiating

between normal users and administrators. The admin interface allows for the control of system content, ensuring that the system is both functional and maintained.

Overall, the results show that the core system features were successfully implemented and work as expected within the project's scope.

### **6.2.2 Testing Outcomes**

Testing was conducted to ensure that the system was both functional and reliable. A combination of white box testing, integration testing and usability testing was used to evaluate the system from both technical and user perspectives, ensuring that it aligned with the functional and non-functional requirements defined earlier in the project.

White box testing confirmed that the system's core logic functions correctly. Core backend processes, such as database insertion, retrieval, and relational integrity, were thoroughly evaluated. Several issues were discovered during development, particularly those with foreign key restrictions and database consistency. These concerns were rectified by refining the database design and improving query structure, resulting in a more robust and dependable data model. This shows that the underlying system logic is reliable and capable of ensuring data integrity.

Black box testing confirmed that the system performs as intended from the user's perspective. Key functionalities such as user authentication, resource upload, search functionality, and flashcard creation were tested with predetermined inputs and expected outputs. The findings showed that the system consistently provides the proper and anticipated outputs for valid inputs. Minor edge-case concerns, such as incomplete form inputs or incorrect data entries, were discovered during testing and resolved with validation enhancements. This demonstrates that the system satisfies its functional requirements from an external perspective.

Integration testing showed that the various system components operate cohesively. Interactions between the presentation layer, application layer and database layers were evaluated to ensure that data flowed seamlessly. For example, resources contributed through the interface were properly stored in the database and then retrieved and shown to the user. Similarly, flashcards were effectively connected to their respective sources, demonstrating the efficacy of the relational database architecture. These results show that the three-tier design works well as a unified system. Usability testing, although limited in scale, indicated that the system is usually intuitive and simple to use. Users found it easy to execute crucial activities including looking for information and generating flashcards. Feedback indicated that the integrated design decreased the need to transition between several tools, increasing workflow productivity and lowering cognitive strain. This reinforces the usability goals established throughout the requirements and design phases.

Overall, the testing results demonstrate that the system is functional, reliable and serves its intended purpose. The combination of different testing methodologies ensures that both internal logic and user-facing features work as expected.

### **6.2.3 Project Planning: Initial and Revised Gantt Chart**

At the start of the project, a Gantt chart was created to outline the expected timeframe and development structure. As illustrated in Appendix A (a.2), the original chart depicts a systematic and mainly sequential workflow that starts with requirements analysis and progresses through design, implementation, testing and assessment. Each step was assigned a fixed period, indicating a standard linear development approach. This approach offered a clear picture of task dependencies and ensured that all key objectives outlined in the proposal were completed within the project period. However, during the implementation phase, it became clear that the development process did not follow a precise linear progression. Instead, activities like database design, interface development and testing were redone several times when new requirements, technical issues and usability improvements emerged. This repetitive activity is not completely captured by the original Gantt chart, which implies one directional flow between phases.

To better depict the real development process, a redesigned Gantt chart was created, as illustrated in Appendix A (a.3). The redesigned chart depicts a more realistic workflow, with actions overlapping and evolving over the project's lifespan. For example, testing activities are depicted to occur concurrently with development rather than as a distinct final step, and implementation tasks are spread across numerous time periods to demonstrate continual improvement. The redesigned Gantt chart was used to track development progress informally, using Trello board divided into sprint-based activities see Figure 1. The board depicts the division of work into certain development phases, such as design, key feature implantation and testing. Although the Trello board shows multiple late tasks with red deadline indications, this is not a realistic representation of project performance. Many of the sprint targets were fulfilled within the specified timeframes however the board was not routinely updated. This demonstrates a limitation in project tracking, rather than execution. Overall, the comparison of the initial and amended Gantt charts demonstrates a crucial project outcome. The shift from a theoretical, plan driven approach to a more flexible and iterative development method. This indicates a better awareness of real-world software development techniques and emphasises the significance of flexibility in managing complex systems.

### **6.2.4 Conclusion**

The results show that the project was successful in creating a functioning prototype that meets many of the identified functional and non-functional requirements. The system combines resource management and flashcard-based revision on a single platform, addressing the fragmentation found in both the



literature review and the product analysis. Testing results demonstrate that the system performs consistently across its primary features, and the new project plan emphasises the necessity of iterative and flexible development techniques. These findings give solid evidence that the system meets its primary goal of enhancing organisation and promoting active recall in an integrated context.

## **6.3 Evaluation**

Following the presentation of the results, this part gives critical evaluation of the projects outcomes in relation to the initial aims, objectives and the larger problem domain. The project's primary aim was to develop an integrated system that included a research repository with a flashcard-based learning tool to help students organise academic materials and retain key information. The evaluation demonstrates that this aim has been generally met, however several limitations and areas for further development have been identified.

From a functional perspective, the system successfully implements the core requirements established during the design process. These include the ability to store and manage academic sources, show structured metadata such as title, author, abstract, and keywords and create flashcards from source content. The system's combination of repository and flashcard functionalities is a crucial strength, allowing users to switch seamlessly from interacting with academic materials to practicing active recall. This directly addresses the issue of fragmented study workflows, in which students frequently use various disconnected tools. When considered within the wider problem, the system reflects current student practices, where platforms are employed to assist in the various aspects of learning. Anki and Quizlet are popular tools for flashcard-based revision and retrieval practice. While these technologies are useful for memorisation, they need users to manually enter information, which may be time-consuming and separated from the original learning material. Platforms for storing and arranging academic information, such as Google Drive and Microsoft OneNote, do not actively encourage revision or knowledge retention. The proposed solution aims to close the gap by combining organisational and learning functionalities into a single platform. By linking flashcards directly to academic sources, the system lowers the need for manual knowledge transfer and allows for a more unified study experience. This provides a significant improvement in process efficiency over conventional fragmented systems. Despite this benefit, the system does not yet have the depth of functionality seen in established tools. Anki, for example, offers powerful spaced repetition algorithms as well as performance tracking, whereas OneNote allows customisable notetaking and multimedia functionality. These functionalities are not completely implemented in the existing system, indicating a need for additional development.

### **6.3.1 Evaluation Criteria/Metrics**

To ensure a systematic review, the system was assessed using a variety of criteria, including functionality, usability, performance, and learning efficacy. Functionality was assessed based on how well the system met its specified requirements. The results indicate that the core features, such as source management, relational database structures, and flashcard generation, were effectively implemented. Heuristic assessment and scenario-based testing were used to assess usability. The UI is intended to be clear and intuitive, with easy navigation between sources, flashcards and question-based features. Elements like structured layouts, keyword displays, and the option to open-source material in a separate tab contributes to the overall user experience. However, without empirical user input, usability judgements are more informative than definitive. During testing, performance was measured in terms of system responsiveness and efficiency. The system functions effectively on a small scale, with quick data retrieval and low latency while creating flashcards. However, scalability has not been properly evaluated, and performance with bigger datasets or multiple users remain uncertain. Learning efficacy was evaluated conceptually, using proven educational principles like active recall. The integration of flashcards with source information promotes retrieval practice, which has been shown to improve knowledge retention. However, the lack of sophisticated capabilities such as spaced repetition restricts the system's potential usefulness when compared to dedicated learning tools.

### **6.3.2 Analysis of results and Objective Achievement**

The results demonstrate that the bulk of the project's objectives have been achieved. The system seamlessly incorporates critical components such as a structured repository, flashcard generation and user interaction features including question-and-answer functionality. These parts work together to serve the overall aim of enhancing how students organise and interact with academic content. However, certain objectives were only partially met. While flashcards are created from source abstracts, the method is quite simple and does not use complex techniques like natural language processing or AI-driven summarisation. This reduces the depth and quality of the resulting flashcards. Furthermore, features like customised learning helps progress monitoring and adaptive revision tactics were not completely implemented. The project also emphasises the challenges involved with early planning and system development. The initial timeframe overestimated the complexity of the database architecture, particularly in terms of many to many relationships and data integrity requirements. As a result, more time was needed for debugging and refining, resulting in a more iterative development process than originally envisioned. This reflects the actual realities of software development, where flexibility and adaptability are required.

### **6.3.3 Limitations and Challenges**

Several limitations were identified during the evaluation procedure. One significant issue is the lack of actual user testing. Due to time and knowledge restraints, the system was tested using hypothetical

user scenarios rather than actual investigations. While this approach provides some light on usability, it does not fully capture user behaviour, happiness or efficacy in real world contexts.

To address this, we evaluated representative user personas. For example, a typical undergraduate student handling many modules would benefit from the system's capacity to arrange sources and create revision materials in one location. Similarly, a student studying for an exam may benefit from the direct relationship between reading materials and flashcards. Appendix E has persona scenarios based on limitations and benefits.

### **6.3.4 Implications and Comparison with Existing Work**

The project's outcomes have a significant implication for the use of digital learning tools. The system shows how merging a research repository with a flashcard-based learning tool may eliminate fragmentation in students' workflows and promote more efficient study practices. This is a significant contribution to the issue area as previous solutions generally target either organisation or revision but not both. Compared to frequently used tools, the system provides a distinct set of characteristics.

Unlike Anki and Quizlet, it connects flashcards to academic materials, reducing manual effort while increasing contextual learning. Unlike Google Drive and OneNote, it has an active learning component that promotes information retention rather than passive storage. However, in other places the system remains less advanced than these established platforms. It lacks features collaborative tools and intelligent recommendations, which are becoming more widespread in the current educational technologies. This demonstrates the difference between prototype development and fully completed commercial solutions. Despite its limitations, the project highlights the possibility and potential benefit of combining repository and flashcard functionality. It lays the foundation for future growth and identifies possibilities to improve digital learning tools through improved integration and automation.

## **6.4 Conclusion**

This chapter highlighted the created system's findings and gave a thorough evaluation of its performance about the original aims and objectives. The results show that the system effectively executes the basic functionality of an integrated research library and flashcard-based learning tool. Key features including resource management, flashcard creation, and contextual navigation were created and tested, demonstrating that the system can support both academic organisation and active recall. The evaluation shows that the system efficiently tackles the issue of fragmented study processes, which was noted in previous chapters. By merging features that are often scattered across several platforms, the system lowers the need for users to transition between tools and promotes a more consistent learning experience.

Overall, the findings and evaluation show that the project made a significant contribution to the issue area by proving the potential of combining repository and flashcard functionality into a single platform. The insights gathered from this chapter serve as a solid basis for the next chapter, which will delve into the system's larger legal, social, ethical, and professional implications.

## **7 LEGAL, SOCIAL, ETHICAL AND PROFESSIONAL ISSUES**

### **7.1 Legal:**

This project must comply with all legal laws like data protection and intellectual property rules and regulations, particularly when dealing with user-generated academic work. According to Liu and Khalil (2023), compliance with the General Data Protection Regulation is critical for systems that gather or handle academic data since it provides openness and permission. According to Welford (2025), GDPR establishes concepts such as lawful processing, data minimisation, and user control over personal information, which is critical for ensuring trust and compliance in digital academic systems. Similarly, the UK Government (2018) specifies how the Data Protection Act 2018 must maintain personal data and is protected under UK law, supplementing the GDPR's framework for responsible data management. As a result, the repository will incorporate privacy warnings and only collect data required for academic reasons. Copyright compliance will also be maintained by allowing users to contribute properly referenced academic writings that must be authorised by an administrator, preventing legal violation of published information.

Within the context of this project, legal constraints have a direct impact on the system design and implementation. The system collects only required user information, such as login credentials, and uses secure handling techniques like password hashing and limited access. Privacy issues are incorporated into the design through minimum data storage and explicit system limits for user information. Furthermore, copyright compliance is addressed by demanding that any academic resources added to the system be properly cited and authorised by an administrator. This decreases the possibility of unlawful dissemination of intellectual information and guarantees that the system runs within legal limits.

### **7.2 Social:**

From a social aspect, the system encourages students to collaborate and share their experiences by also requesting permission for specific sources to be added. As highlighted by Khalil et al. (2023), educational technologies can enhance peer interaction and motivation when developed with inclusiveness with accessibility in mind. This system supports social equality by being web-based and accessible across numerous platforms, eliminating digital barriers for students from diverse backgrounds, and including accessibility features like as screen reader compatibility, configurable text size, and clear visual layouts. These design considerations enable students with visual or reading

challenges to interact with academic information more freely and efficiently, ensuring that the platform is accessible to all users.

In terms of the entire project, these societal factors influenced important design decisions, especially the emphasis on simplicity and accessibility. The system was intentionally developed as a web-based platform to ensure ease of access across devices, reducing dependency on specific technologies. Furthermore, by combining repository and flashcard functionality into a single platform, the solution eliminates the need for various tools, reducing cognitive load and promoting a more inclusive learning experience for students with diverse study preferences and requirements.

### **7.3 Ethical:**

This project prioritises academic integrity, fairness and ethical technology usage in an educational environment. According to Huang et al. (2025), ensuring transparency, privacy and accountability in digital education platforms is critical for preventing unethical activity and increasing user confidence. In compliance with these findings, the proposed repository would guarantee that all user data is anonymised and handled appropriately, while also including plagiarism prevention measures and citation prompts to promote academic integrity. Furthermore, the system's design will ensure algorithmic fairness by preventing bias in flashcard production or source evaluation while promoting equitable learning chances for all users.

These ethical issues were integrated into the system development process rather than applied at a single stage. For example, connecting flashcards to their original academic sources promotes responsible information usage and lowers the danger of decontextualised or misleading learning. Furthermore, safe data processing techniques ensure that user information is secured, which increases trust in the system. This indicates that ethical responsibility is not only a theoretical concern, but an active part of system design and execution.

### **7.4 Professional:**

From a professional viewpoint, this project adheres to recognised computing standards for integrity, accountability, and ethical responsibility. Insights from O'Shea (2024), emphasise the importance of incorporating ethics and professional conduct into computing curricula, reinforcing the need for projects like this to adhere to ethical best practices from the start. Additionally, Blaise et al. (2024) highlights that well-structured ethical frameworks assist information systems professionals in

navigating complicated problems such as privacy, data protection, and responsible system design. By incorporating these principles into the building of the system, including secure processing of academic data, fair depiction of material, and accessible design, the project displays commitment to professional standards and assures responsibility.

From a development standpoint, these professional standards are represented in the technological implementation of the system. The use of structured code organisation, safe database interactions using prepared statements, and role-based access control indicates compliance with software engineering best practices. Furthermore, the modular system architecture promotes maintainability and future scalability, which are critical factors of professional development. This guarantees that the system is not only functional but also designed to meet industry requirements.

## **7.5 Conclusion**

In conclusion, legal, social, ethical, and professional factors influenced the design and development of this system. Compliance with data protection laws and copyright restrictions ensures that the system runs legally, while social concerns support accessibility and inclusion for a wide range of users. To ensure appropriate use of technology in education, the system incorporates ethical concepts such as privacy, equality, and academic integrity. At the same time, professional standards governed the development process, guaranteeing that the system is safe, maintainable, and follows software engineering best practices. These considerations work together to strengthen the system's credibility and reliability, indicating that it is not just technically functioning but also socially responsible and ethically sound. This lays a solid framework for the last chapter, which examines the overall project results and potential development prospects.

## 8 CONCLUSION

### 8.1 Summary of the Project

This project sought to address a frequent challenge experienced by university students, the difficulty of managing vast amounts of academic material while efficiently retaining key information. The aim was to create a web-based system that combines a research repository with a flashcard-based learning tool, allowing users to manage materials and apply active recall on a single platform.

To achieve this, the project followed a structured and research informed approach. The literature review identified many key issues, including information overload, fragmented study processes, and a lack of integration between organisational tools and revision systems. A product evaluation further reinforced these findings by highlighting the strengths and limitations of commonly used tools, including Google Drive, Microsoft OneNote, Anki and Quizlet.

These insights influenced the requirements analysis and system design, which were created using UML modelling, ER Diagrams and user interface prototypes. The system was built using PHP, HTML and MySQL with a modular structure and three-tier architecture. Core functionality including resource management, flashcard creation, contextual linking, and user authentication have all been successfully developed and tested. The results demonstrated that the system is functional and capable of supporting both academic organisation and active recall.

### 8.2 Evaluation of Outcomes and Objectives

The overall findings indicate that the project has successfully achieved its primary aim. The developed system demonstrates that it is feasible to integrate a research repository with a flashcard learning tool in a way that enhances workflow efficiency and promotes more productive study habits. By decreasing the need to transition between several application, the system addresses the fragmentation issue identified in the literature and product review. The evaluation also demonstrates that the system is compatible with evidence-based learning practices, particularly retrieval practice. The ability to generate flashcards from academic sources and link them back to the original content allows for a more organised and context-aware revision process. This is a significant improvement over typical procedures, in which students manually transit data between separate tools.

Although most objectives were met, several restrictions remain. The system still lacks sophisticated features like spaced repetition algorithms, personalised learning analytics and AI-driven content generation. Furthermore, the absence of large-scale user testing implies that findings about usability and learning efficacy are dependent on theoretical evaluation rather than empirical evidence. These



limitations highlight the difference between a functional prototype and a fully developed system. From a broader perspective, the project contributes to the field of digital learning tools by demonstrating the value of integration. It builds upon existing systems by combining their strengths while addressing their main limitations. Although it does not replace existing platforms, it provides a foundation for future systems that aim to integrate organisation and learning within a single environment.

### **8.3 Reflection on the Project and Learning**

The development of this project provided valuable insights into both technical implementation and the broader software development process. One of the most important lessons learnt was the complexity of database architecture, especially when implementing connections like many-to-many mappings and ensuring data integrity with foreign key restrictions. These challenges required continual refinement and demonstrated the value of detailed planning and testing. The implementation of an Agile-inspired development approach was beneficial, allowing the system to evolve gradually and adapt to new challenges as they emerged. This iterative process allowed us to optimise both the system's design and functionality, but it also demonstrated how difficult it is to precisely estimate development time during the initial planning phase. From a design perspective, the project emphasised the value of simplicity and functionality. Addressing the issue of cognitive overload required ensuring that the system remained simple and easy to navigate. Furthermore, integrating design decisions to literature results helped guarantee that the system was not just technically functional but also based on educational theory.

Overall, the project improved understanding of full-stack web development, database administration, and user-centred design. It has also illustrated the significance of matching technological solutions to real-world challenges and user requirements.

### **8.4 Future Work**

While the system that was developed has a solid foundation, there are various areas for future development and expansion.

One important area for improvement is the implementation of spaced repetition algorithms similar to those used in Anki. This would enable the system to automatically schedule flashcard evaluations based on user performance, considerably increasing its usefulness as a learning tool. Furthermore, implementing AI-driven features for automatic flashcard production may increase the quality and depth of revision material while minimising the need for manual input.

Another potential enhancement is the implementation of user metrics and progress monitoring. Performance dashboards, revision data, and tailored suggestions might help users better understand their learning patterns and progress. This would bring the system into close alignment with contemporary instructional technology. Scalability and deployment are also essential topics for future work. To evaluate performance under real-world situations, the system would need to be tested with larger datasets and several concurrent users. Further development might involve cloud deployment, which would allow the system to be accessible more broadly. Furthermore, rigorous user testing should be carried out to ensure usability and efficacy. Collecting input from a varied set of students would give significant insights into how the system is utilised in practice while also identifying opportunities for development. This would strengthen the evaluation and allow for additional system refinement.

Finally, this project illustrates the feasibility of merging academic resource management and active recall approaches on a single platform. While the current system is a functional prototype, it also outlines a clear route for future growth. The work adds to the continuing effort to improve digital learning environments and emphasises the necessity of building systems that facilitate both organisation and meaningful engagement with academic information.

## REFERENCES

- Arden, B., Norris, J., Cole, S. and Gamble, S.C. (2024). Digital notetaking in lectures: how students adapt to a multi-faceted university learning environment. *Cogent education*, 11(1). doi:<https://doi.org/10.1080/2331186x.2024.2373552>
- Blaise, O.O., Aaron, I., Alfred, U. and Afolarin Amusa (2024). Evaluating the Ethical Frameworks of Information Security Professionals: A Comparative Analysis. *Asian Journal of Computer Science and Technology*, 13(2), pp.61–66. doi:<https://doi.org/10.70112/ajcst-2024.13.2.4289>
- Bond, M., Marín, V.I., Dolch, C., Bedenlier, S. and Zawacki-Richter, O. (2018). Digital transformation in German higher education: student and teacher perceptions and usage of digital media. *International Journal of Educational Technology in Higher Education*, 15(1). doi:<https://doi.org/10.1186/s41239-018-0130-1>.
- Carpenter, S.K., Pan, S.C. and Butler, A.C. (2022). The science of effective learning with spacing and retrieval practice. *Nature Reviews Psychology*, [online] 1(9), pp.496–511. doi:<https://doi.org/10.1038/s44159-022-00089-1>.
- Firth, J., Rivers, I. and Boyle, J. (2021). A Systematic Review of Interleaving as a Concept Learning Strategy. *Review of Education*, [online] 9(2). doi:<https://doi.org/10.1002/rev3.3266>.
- Friess, E. (2022). Scrum in Classroom Collaborations: A Quasi-Experimental Study. *Journal of Business and Technical Communication*, p.105065192211218. doi:<https://doi.org/10.1177/10506519221121817>
- Huang, C.L., Shao, X., Wu, C. and Yang, S.C. (2025). Navigating the digital learning landscape: insights into ethical dilemmas and academic misconduct among university students. *International Journal of Educational Technology in Higher Education*, 22(1). doi:<https://doi.org/10.1186/s41239-025-00516-2>
- Khalil, M., Slade, S. and Prinsloo, P. (2023). Learning analytics in support of inclusiveness and disabled students: a systematic review. *Journal of Computing in Higher Education*, [online] 36. doi:<https://doi.org/10.1007/s12528-023-09363-4>
- Liu, Q. and Khalil, M. (2023). Understanding privacy and data protection issues in learning analytics using a systematic review. *British Journal of Educational Technology*, 54(6), pp.1715–1747. doi:<https://doi.org/10.1111/bjet.13388>

- Lu, M., Farhat, J.H. and Beck Dallaghan, G.L. (2021). Enhanced Learning and Retention of Medical Knowledge Using the Mobile Flash card Application Anki. *Medical Science Educator*, [online] 31(6), pp.1975–1981. doi:<https://doi.org/10.1007/s40670-021-01386-9>
- Nielsen, J. (1994). *10 Heuristics for User Interface Design*. [online] Nielsen Norman Group. Available at: <https://www.nngroup.com/articles/ten-usability-heuristics/>
- O’Shea, K. (2024). Evaluating UK academics’ perspectives of ethics education within computer science degree programmes: a preliminary insight. *International Journal of Ethics Education*. doi:<https://doi.org/10.1007/s40889-024-00205-0>
- Santos, M.F., Filipe, R.A. and Cunha, J.C. (2024). From Traditional to Agile Methodologies in Software Project Management Education: A Case Study. *From Traditional to Agile Methodologies in Software Project Management Education*, pp.491–497. doi:<https://doi.org/10.1145/3702163.3702460>
- Sillence, E., Dawson, J.A., McKellar, K. and Neave, N. (2022). How do students use digital technology to manage their university-based data: strategies, accumulation difficulties and feelings of overload? *Behaviour & Information Technology*, 42(14), pp.1– 10. doi:<https://doi.org/10.1080/0144929x.2022.2126948>
- Sun, M., Tsai, S., Engle, D.L. and Holmer, S. (2021). Spaced Repetition Flashcards for Teaching Medical Students Psychiatry. *Medical Science Educator*, 31(3). doi:<https://doi.org/10.1007/s40670-021-01286-y>.
- Trumble, E., Lodge, J.M., Mandrusiak, A. and Forbes, R. (2023). Systematic review of distributed practice and retrieval practice in health professions education. *Advances in Health Sciences Education*, 29, pp.689–714. doi:<https://doi.org/10.1007/s10459-023-10274-3>.
- UK Government (2018). *Data Protection Act*. [online] Gov.uk. Available at: <https://www.gov.uk/data-protection> [Accessed 15 Oct. 2025].
- van Wijk, E.V., de Jonge, M., van Blankenstein, F.M., Janse, R.J. and Langers, A.M.J. (2024). The battle of question formats: a comparative study of retrieval practice using very short answer questions and multiple choice questions. *BMC Medical Education*, 24(1). doi:<https://doi.org/10.1186/s12909-024-06538-0>.
- Varkey Nadakkavukaran Santhosh, Coutinho, D., Ankola, A.V., Yuvarani Kandasamy Parimala, Siva Shankkari and Kavitha Ragu (2024). Effectiveness of spaced repetition learning using a mobile flashcard application among dental students: A randomized controlled trial. *Journal of dental education*. doi:<https://doi.org/10.1002/jdd.13561>
- Wolford, B. (2025). *What is GDPR, the eu’s new data protection law?* [online] GDPR.EU. Available at: <https://gdpr.eu/what-is-gdpr/> [Accessed 15 Oct. 2025].

# APPENDIX A - Project Proposal

## a.1 Original Proposal

COMP1682 Project Proposal

Integrated Study and Research Repository with Flashcard-Based Learning

Richard Egenti

BSc H COMPUTING (EXT)

001239105

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## INTRODUCTION/OVERVIEW

University students are increasingly challenged by dual demands of managing large volumes of academic material and retaining key information for long term use. As higher education shifts to digital learning environments, the volumes of notes have grown substantially through articles, blogs, books and more being available online, this has often led to disorganisation and information overload for higher education students. Sillence et al. (2022) found that while students have found various ways to manage their university-based data there are still many who experience difficulties with information overload. Similarly, Arden et al. (2024) highlights how students adapt to digital note-taking practices in lectures but also face different challenges trying to effectively integrate them into the learning routines. These findings suggest that there is a need for more effective systems that support both resource management and effective study practices.

Also, research in educational psychology consistently demonstrates that practices based on spaced repetition are among the most successful practices for improving long term retention. Varkey et al. (2024) showed that a mobile flashcard application consisting of spaced repetition significantly enhanced dental students' long-term knowledge compared to more traditional methods. Similarly, Lu et al. (2021) reported that the use of the flashcard application Anki was effective for better learning efficiency and stronger in-depth medical knowledge.

This project attempts to help higher education students overcome two typical challenges - knowledge retention and information overload. It will accomplish this by creating a web-based program that combines a research repository and a flashcard learning system. Academic materials saved in the database may be accessed directly via the platform, and each source will be connected to automatically created flashcards that emphasise key points or subjects. When students examine these flashcards, selecting a certain point directs them to the matching section of the original source. This study aims to improve the efficiency and engagement of students in organising, accessing, and retaining academic material by integrating systematic resource management with active recall.

## AIM

The aim of this project is to develop a web-based application that combines a research library with flashcards to assist students in organising academic resources, extracting key concepts, and practicing active recall. Therefore, improving information retention, reducing overload, and encouraging better study habits.

## OBJECTIVES

### Analysis and Requirements Gathering

Research into existing research repository tools and flashcard systems, evaluating their functionality and limitations through academic sources. [5.0]

Write a literature review on the importance of information management systems and active recall methods in higher education, with clear analysis of assumptions and limitations. [12.0]

Research functional and non-functional requirements for an integrated repository and flashcard system, based on insights from the literature review and user expectations. [3.0]

Verify requirements by designing and distributing a questionnaire or short survey to students on their study habits and tool preferences. [5.0]

### Design

Create a rich picture diagram showing students, their study workflows, and how repository and flashcard features interact. [3.0]

Create use-case diagrams illustrating system functionalities such as uploading resources, generating flashcards, and reviewing content. [3.0]

Create a conceptual ERD to outline the entities and their relationships. [3.0]

Create a physical ERD diagram specifying attributes, keys, and database schema structure. [6.0]

Design possible user interfaces using tools such as Figma or Axure RP. [7.0]

### Development and Implementation

Implement the database schema by creating tables and relationships in MySQL or MongoDB. [10.0]

Develop the main user interface using a web framework using PHP and Html. [13.0]

Implement the resource upload and organisation features, allowing users to add, tag, and search academic sources. [8.0]

Implement flashcard generation linked to stored resources, displaying key concepts or topics. [13.0]

Create functionality where clicking a flashcard topic redirects users to the relevant section in the source. [8.0]

Implement user registration and login functionality with secure authentication. [8.0]

Add an admin panel to manage users, resources, and system data. [8.0]

### **Testing**

Usability testing to observe how easily students navigate the repository and flashcard features. [5.0]

Black box testing to ensure system functions work as expected for users unfamiliar with the code. [5.0]

White box testing to verify system logic and database interactions for flashcard generation and resource retrieval. [5.0]

### **Evaluation**

Evaluate whether the system meets the initial functional and non-functional requirements. [5.0]

Assess the system's usefulness in reducing information overload and improving knowledge retention through small-scale user feedback. [5.0]

Discuss limitations, challenges, and potential improvements for future development. [5.0]

## **PROJECT FRAMEWORK OR ANY METHODOLOGY USED**

The Agile methodology is adopted for this project because it supports a user-friendly approach and an extremely flexible approach to system development, which is essential for building the learning platform that integrates a research repository and flashcards. The process enables the project to move through organised but adaptive stages of design, implementation, testing, and evaluation, guaranteeing that each system component is constantly modified in response to user feedback.

During the design stage, Agile facilitates the construction of user-centered interface prototypes, such as the repository dashboard, flashcard viewer, and source-linking functions, which are jointly built to meet student demands. The implementation and development stages are divided into small, manageable sprints, allowing for incremental improvements and rapid adjustments depending on testing findings. Although this project is being created independently, Scrum concepts will be modified for solo usage by considering each sprint as a concentrated development period with well-defined goals, such as design, implementation, and testing. Progress will be tracked via self-review at the conclusion of each sprint, allowing for reflection, refinement, and job prioritisation in the following phase.



Frequent testing ensures that the interface stays intuitive and that data retrieval and flashcard creation work properly. Finally, the evaluation step compares the project outputs to the original objectives, focusing on usability, functionality, and improved knowledge retention.

According to Santos et al. (2024), Agile practices such as Scrum use adaptability, collaboration and SMART project outcomes within academic environments. Similarly, Friess (2023) found that groups using Scrum achieved higher efficiency through better communication and better performance scores compared to traditional approaches. These findings support the use of agile for this project, as it allows for continued refinement to ensure that the final product meets students' needs for information organisation and long-term knowledge retention.

## **LEGAL, SOCIAL, ETHICAL AND PROFESSIONAL**

### **Legal**

This project must comply with all legal laws like data protection and intellectual property rules and regulations, particularly when dealing with user-generated academic work. According to Liu and Khalil (2023), compliance with the General Data Protection Regulation is critical for systems that gather or handle academic data since it provides openness and permission. According to Welford (2025), GDPR establishes concepts such as lawful processing, data minimisation, and user control over personal information, which is critical for ensuring trust and compliance in digital academic systems. Similarly, the UK Government (2018) specifies how the Data Protection Act 2018 must maintain personal data and is protected under UK law, supplementing the GDPR's framework for responsible data management. As a result, the repository will incorporate privacy warnings and only collect data required for academic reasons. Copyright compliance will also be maintained by allowing users to contribute properly referenced academic writings that must be authorised by an administrator, preventing legal violation of published information.

### **Social**

From a social aspect, the system encourages students to collaborate and share their experiences by also requesting permission for specific sources to be added. As highlighted by Khalil et al. (2023), educational technologies can enhance peer interaction and motivation when developed with inclusiveness with accessibility in mind. This system supports social equality by being web-based and accessible across numerous platforms, eliminating digital barriers for students from diverse backgrounds, and including accessibility features like as screen reader compatibility, configurable text size, and clear visual layouts. These design considerations enable students with visual or reading

challenges to interact with academic information more freely and efficiently, ensuring that the platform is accessible to all users.

### **Ethical**

This project prioritises academic integrity, fairness and ethical technology usage in an educational environment. According to Huang et al. (2025), ensuring transparency, privacy and accountability in digital education platforms is critical for preventing unethical activity and increasing user confidence. In compliance with these findings, the proposed repository would guarantee that all user data is anonymised and handled appropriately, while also including plagiarism prevention measures and citation prompts to promote academic integrity. Furthermore, the system's design will ensure algorithmic fairness by preventing bias in flashcard production or source evaluation while promoting equitable learning chances for all users.

### **Professional**

From a professional viewpoint, this project adheres to recognised computing standards for integrity, accountability, and ethical responsibility. Insights from O'Shea (2024), emphasise the importance of incorporating ethics and professional conduct into computing curricula, reinforcing the need for projects like this to adhere to ethical best practices from the start. Additionally, Blaise et al. (2024) highlights that well-structured ethical frameworks assist information systems professionals in navigating complicated problems such as privacy, data protection, and responsible system design. By incorporating these principles into the building of the system, including secure processing of academic data, fair depiction of material, and accessible design, the project displays commitment to professional standards and assures responsibility.

## **PLANNING (SEE APPENDIX A & B)**

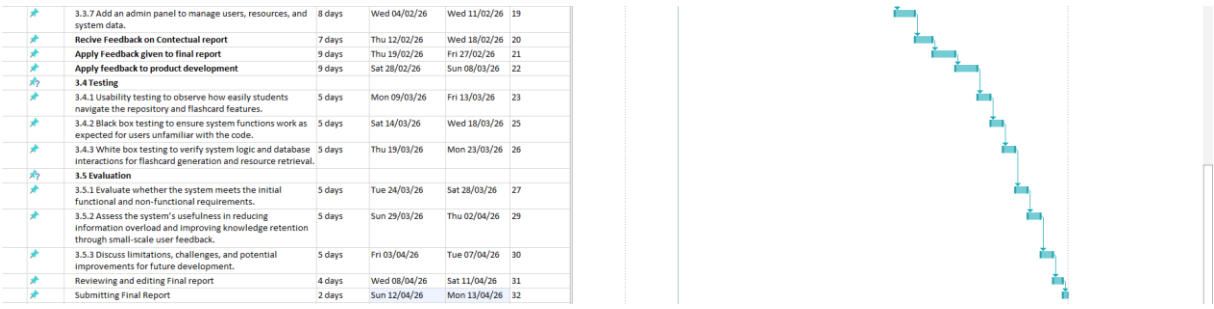
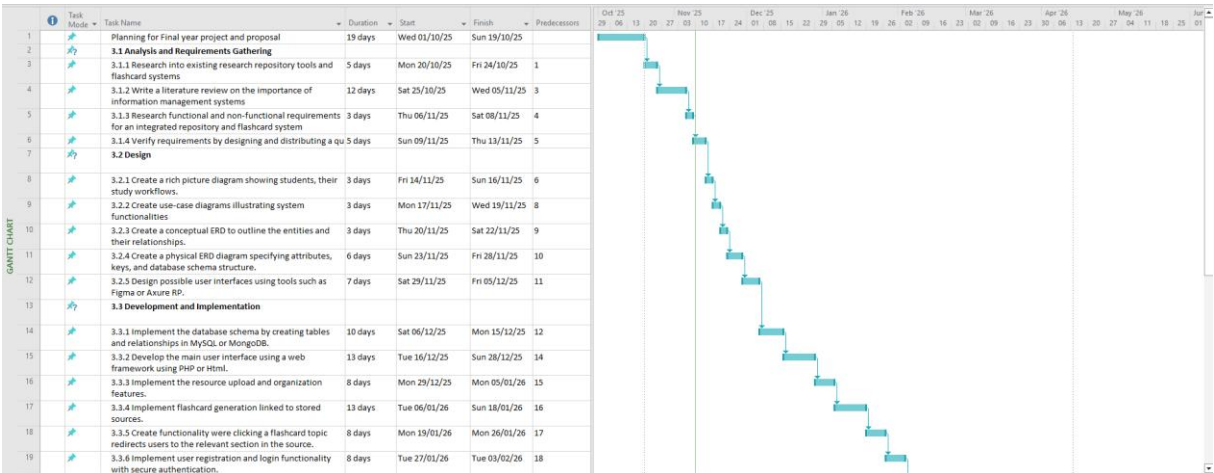
The planning approach starts with gathering and assessing the requirements through research and user feedback, followed by the creation of diagrams and interface prototypes to visualise the repository and flashcard integration. Development will concentrate on implementing the database and core system functions in manageable stages, followed by focused testing such as usability, black box and white box testing to assure reliability and accessibility. The assessment stage will compare the final system against the project requirements and user expectations, determining how well it promotes academic resource organisation and knowledge retention.

The Agile methodology will be used to design the project, which will be completed progressively, allowing for flexibility and refinement along the process (Santos et al., 2024). Each sprint on the Gantt chart corresponds to a critical step of the project, from the analysis, design, development, testing and finally evaluation, with time estimates determined by the objectives. The nature of Agile allows for a regular self-evaluation at the conclusion of each sprint, assuring ongoing progress and compatibility with project objectives. The system will progress via several prototypes that will be adjusted based on the feedback and ensuring that each stage contributes to the final web-based solution.

Tools and Technologies

For this project I intend to use Visual Studio Code for coding with html and php to create the web-based program for my final product and all its functionalities. I will use CSS for the design of the web-based program and have some JavaScript elements for the interface and some of the features of the program. I will use either MongoDB or MySQL to create the back-end database. The design diagrams will be created using Lucidchart which will accommodate the proper creation of objects and their relationships.

APPENDICES



## REFERENCES

- Arden, B., Norris, J., Cole, S. and Gamble, S.C. (2024). Digital notetaking in lectures: how students adapt to a multi-faceted university learning environment. *Cogent education*, 11(1). doi:<https://doi.org/10.1080/2331186x.2024.2373552>
- Blaise, O.O., Aaron, I., Alfred, U. and Afolarin Amusa (2024). Evaluating the Ethical Frameworks of Information Security Professionals: A Comparative Analysis. *Asian Journal of Computer Science and Technology*, 13(2), pp.61–66. doi:<https://doi.org/10.70112/ajcst-2024.13.2.4289>
- Friess, E. (2022). Scrum in Classroom Collaborations: A Quasi-Experimental Study. *Journal of Business and Technical Communication*, p.105065192211218. doi:<https://doi.org/10.1177/10506519221121817>
- Huang, C.L., Shao, X., Wu, C. and Yang, S.C. (2025). Navigating the digital learning landscape: insights into ethical dilemmas and academic misconduct among university students. *International Journal of Educational Technology in Higher Education*, 22(1). doi:<https://doi.org/10.1186/s41239-025-00516-2>
- Khalil, M., Slade, S. and Prinsloo, P. (2023). Learning analytics in support of inclusiveness and disabled students: a systematic review. *Journal of Computing in Higher Education*, [online] 36. doi:<https://doi.org/10.1007/s12528-023-09363-4>
- Liu, Q. and Khalil, M. (2023). Understanding privacy and data protection issues in learning analytics using a systematic review. *British Journal of Educational Technology*, 54(6), pp.1715–1747. doi:<https://doi.org/10.1111/bjet.13388>
- Lu, M., Farhat, J.H. and Beck Dallaghan, G.L. (2021). Enhanced Learning and Retention of Medical Knowledge Using the Mobile Flash card Application Anki. *Medical Science Educator*, [online] 31(6), pp.1975–1981. doi:<https://doi.org/10.1007/s40670-021-01386-9>

O'Shea, K. (2024). Evaluating UK academics' perspectives of ethics education within computer science degree programmes: a preliminary insight. *International Journal of Ethics Education*. doi:<https://doi.org/10.1007/s40889-024-00205-0>

Santos, M.F., Filipe, R.A. and Cunha, J.C. (2024). From Traditional to Agile Methodologies in Software Project Management Education: A Case Study. *From Traditional to Agile Methodologies in Software Project Management Education*, pp.491–497. doi:<https://doi.org/10.1145/3702163.3702460>

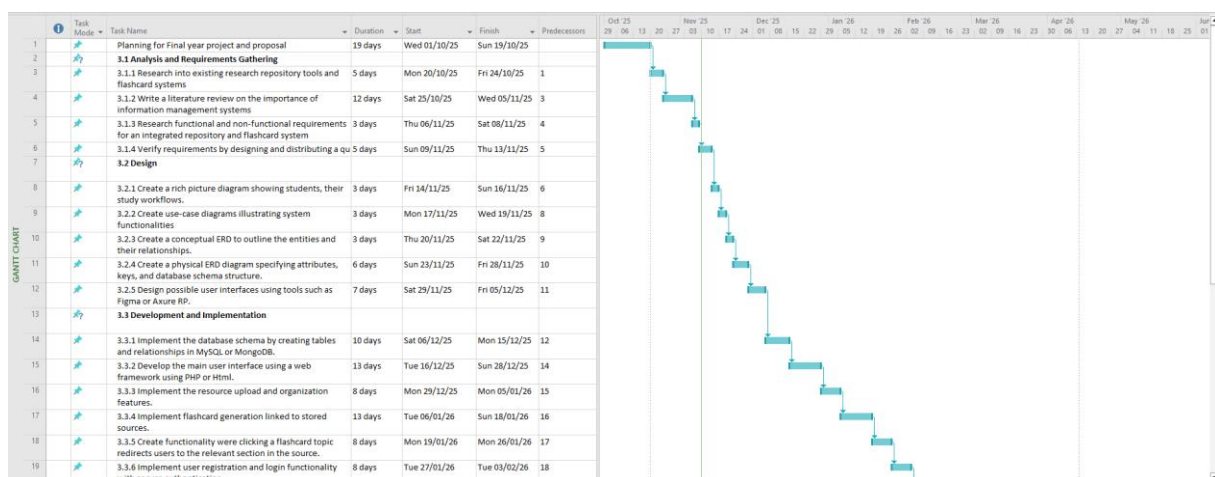
Sillence, E., Dawson, J.A., McKellar, K. and Neave, N. (2022). How do students use digital technology to manage their university-based data: strategies, accumulation difficulties and feelings of overload? *Behaviour & Information Technology*, 42(14), pp.1–10. doi:<https://doi.org/10.1080/0144929x.2022.2126948>

UK Government (2018). *Data Protection Act*. [online] Gov.uk. Available at: <https://www.gov.uk/data-protection> [Accessed 15 Oct. 2025].

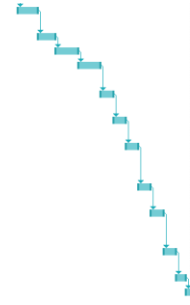
Varkey Nadakkavukaran Santhosh, Coutinho, D., Ankola, A.V., Yuvarani Kandasamy Parimala, Siva Shankkari and Kavitha Ragu (2024). Effectiveness of spaced repetition learning using a mobile flashcard application among dental students: A randomized controlled trial. *Journal of dental education*. doi:<https://doi.org/10.1002/jdd.13561>

Wolford, B. (2025). *What is GDPR, the eu's new data protection law?* [online] GDPR.EU. Available at: <https://gdpr.eu/what-is-gdpr/> [Accessed 15 Oct. 2025].

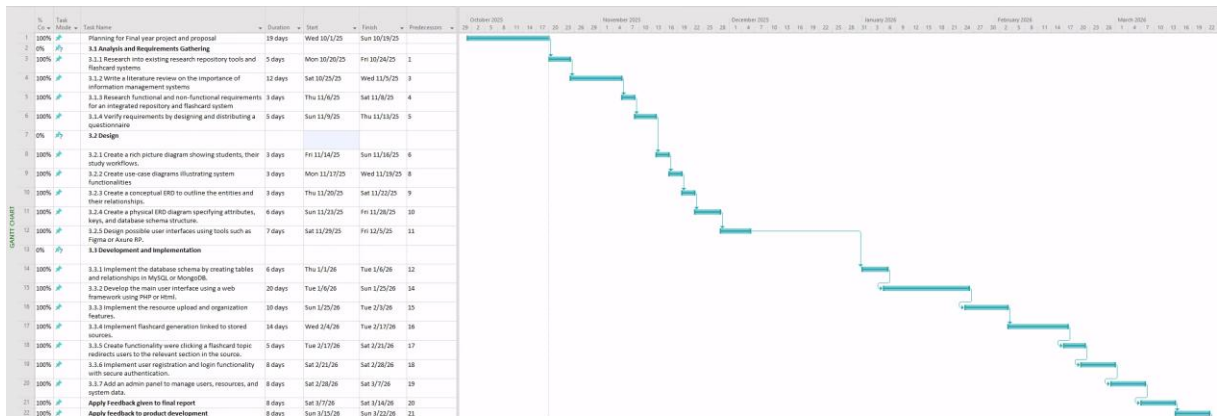
## a.2 Project Plan



|       |                                                                                                                                      |        |              |              |    |
|-------|--------------------------------------------------------------------------------------------------------------------------------------|--------|--------------|--------------|----|
| 3.3.7 | Add an admin panel to manage users, resources, and system data.                                                                      | 8 days | Wed 04/02/26 | Wed 11/02/26 | 19 |
|       | <b>Recive Feedback on Contextual report</b>                                                                                          | 7 days | Thu 12/02/26 | Wed 18/02/26 | 20 |
|       | <b>Apply Feedback given to final report</b>                                                                                          | 9 days | Thu 19/02/26 | Fri 27/02/26 | 21 |
|       | <b>Apply feedback to product development</b>                                                                                         | 9 days | Sat 28/02/26 | Sun 08/03/26 | 22 |
| 3.4   | <b>Testing</b>                                                                                                                       |        |              |              |    |
| 3.4.1 | Usability testing to observe how easily students navigate the repository and flashcard features.                                     | 5 days | Mon 09/03/26 | Fri 13/03/26 | 23 |
| 3.4.2 | Black box testing to ensure system functions work as expected for users unfamiliar with the code.                                    | 5 days | Sat 14/03/26 | Wed 18/03/26 | 25 |
| 3.4.3 | White box testing to verify system logic and database interactions for flashcard generation and resource retrieval.                  | 5 days | Thu 19/03/26 | Mon 23/03/26 | 26 |
| 3.5   | <b>Evaluation</b>                                                                                                                    |        |              |              |    |
| 3.5.1 | Evaluate whether the system meets the initial functional and non-functional requirements.                                            | 5 days | Tue 24/03/26 | Sat 28/03/26 | 27 |
| 3.5.2 | Assess the system's usefulness in reducing information overload and improving knowledge retention through small-scale user feedback. | 5 days | Sun 29/03/26 | Thu 02/04/26 | 29 |
| 3.5.3 | Discuss limitations, challenges, and potential improvements for future development.                                                  | 5 days | Fri 03/04/26 | Tue 07/04/26 | 30 |
|       | <b>Reviewing and editing Final report</b>                                                                                            | 4 days | Wed 08/04/26 | Sat 11/04/26 | 31 |
|       | <b>Submitting Final Report</b>                                                                                                       | 2 days | Sun 12/04/26 | Mon 13/04/26 | 32 |



## a.3 Revised Project Plan



## APPENDIX B - Designs

### b.1 Low Fidelity

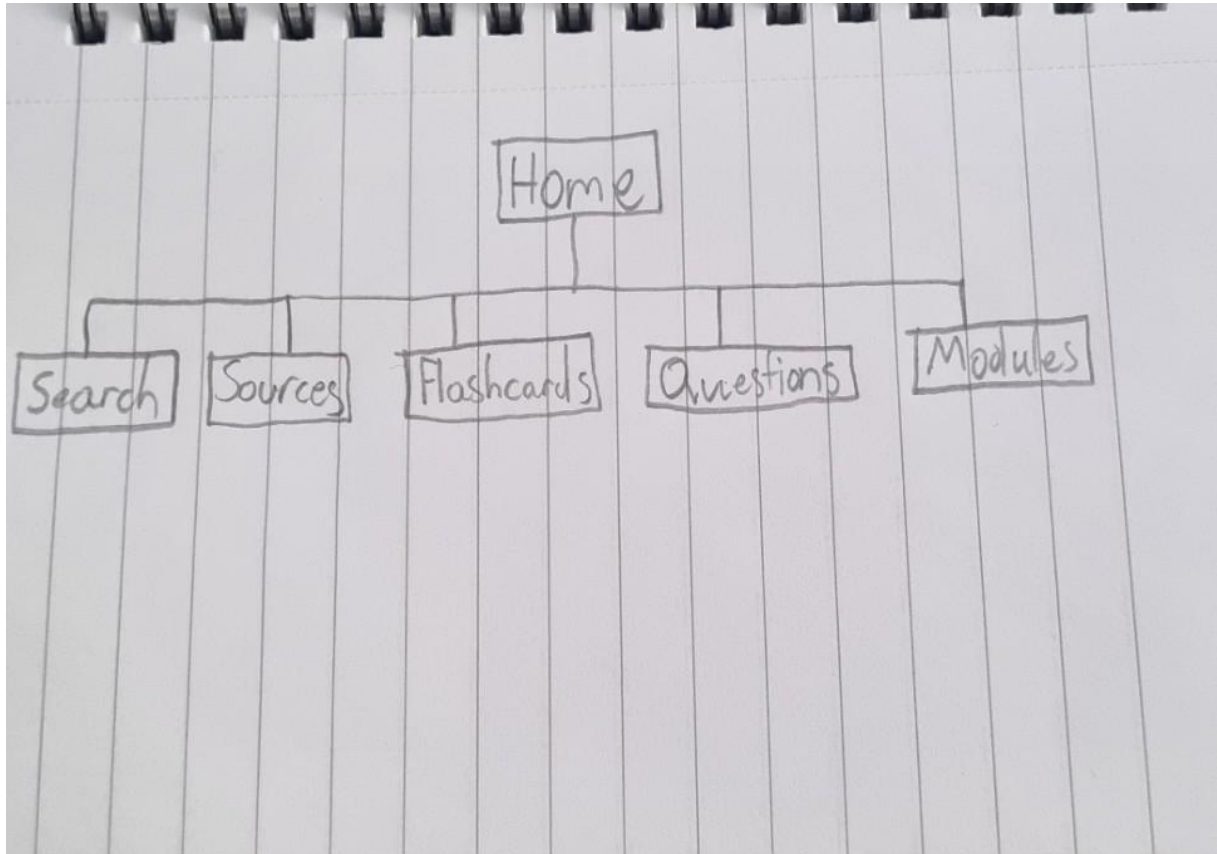


Figure b1.1

Figure b1.1 is a low-fidelity navigation layout for my website.

## b.2 Mid User Fidelity



Figure b2.1

Figure b2.1 is the home page for the website.



Figure b2.2

Figure b2.2 is the sources page where users can see all the sources.



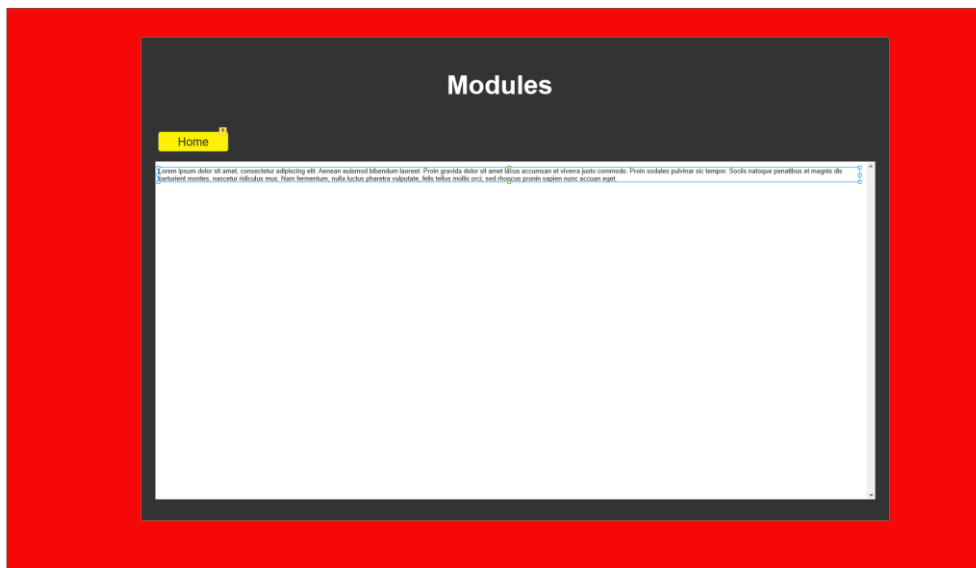


Figure b2.3

Figure b2.3 is the modules page which allows users to see the modules available and the material posted under these modules.

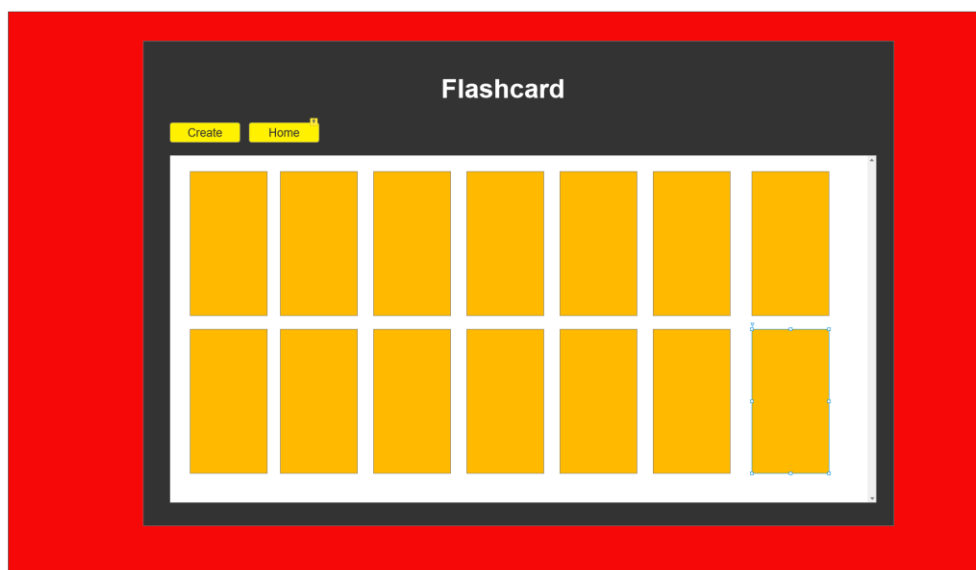


Figure b2.4

Figure b2.4 is the flashcard page which will allow users to create flashcard and use them for revision.

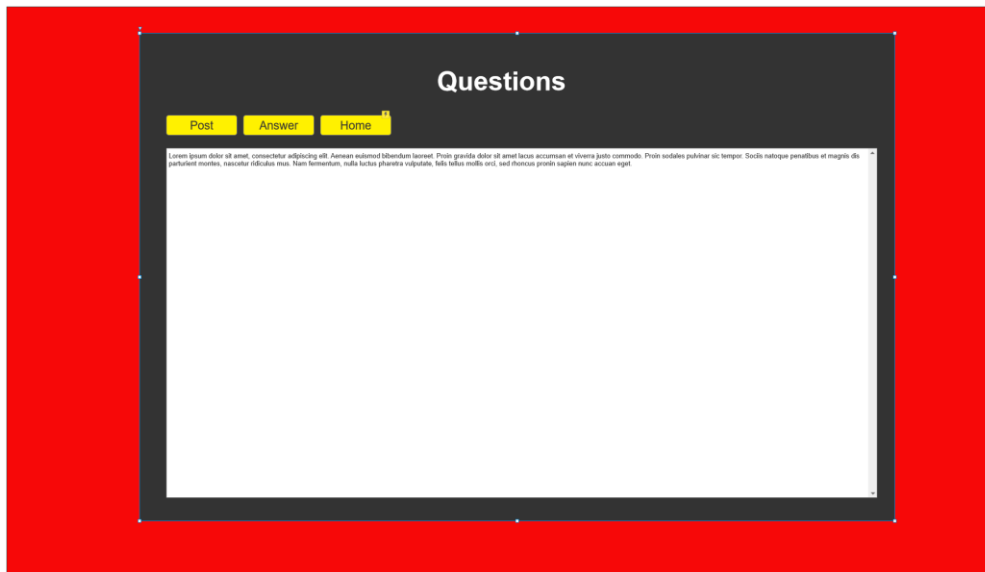


Figure b2.5

Figure b2.5 is the forum page which will allow users to post questions and answer them.

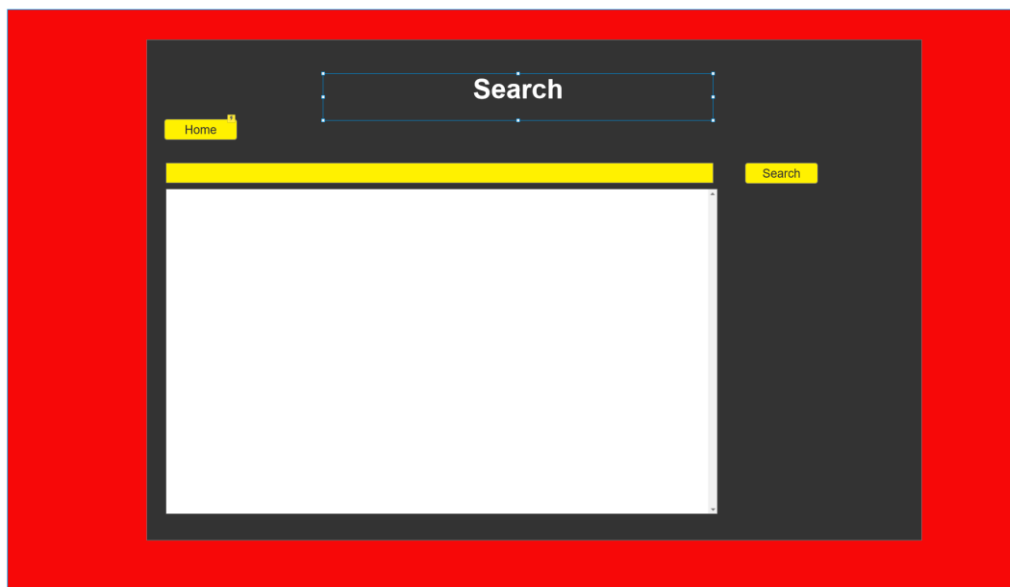
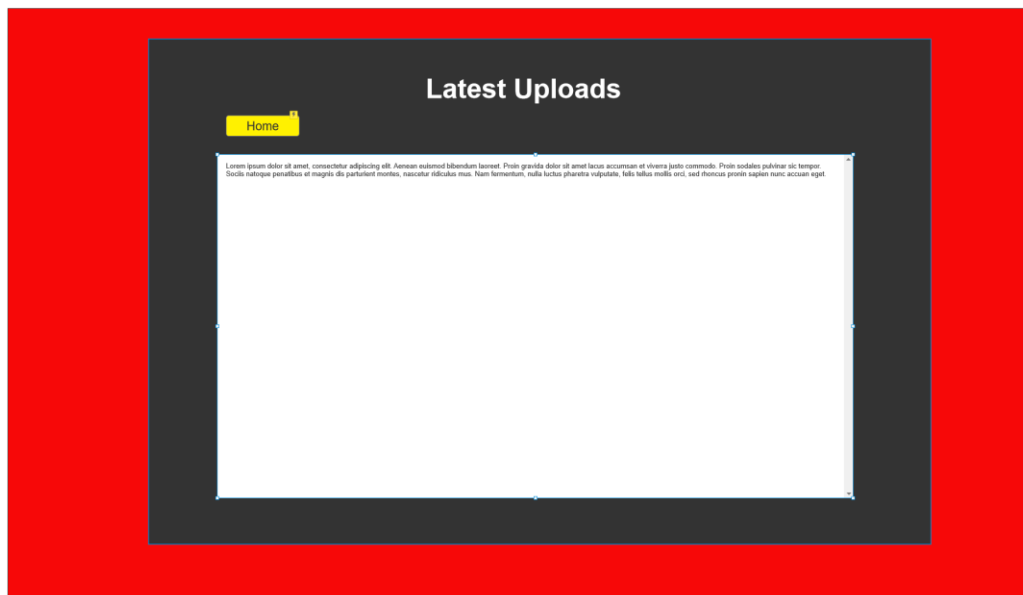


Figure b2.6

Figure b2.6 is the search page that allows users to search for sources and flashcards.



*Figure b2.7*

Figure b2.7 is the page which will show all the latest uploads for sources, flashcards and questions.

## APPENDIX C - Testing Cases

Table 1: User Authentication

| Test ID            | Feature      | Input                  | Expected Output           | Actual Output                             | Result |
|--------------------|--------------|------------------------|---------------------------|-------------------------------------------|--------|
| TC1<br>(Figure E1) | Login        | Valid email + password | User logs in successfully | Login successful, redirected to dashboard | Pass   |
| TC2<br>(Figure E2) | Login        | Invalid password       | Error message displayed   | Error message shown                       | Pass   |
| TC3<br>(Figure E2) | Login        | Empty fields           | Validation error          | Error displayed                           | Pass   |
| TC4<br>(Figure E3) | Registration | Valid details          | Account created           | User added to database                    | Pass   |

Table 2: Resource Management Testing

| Test ID            | Feature         | Input          | Expected Output           | Actual Output               | Result |
|--------------------|-----------------|----------------|---------------------------|-----------------------------|--------|
| TC5<br>(Figure E4) | Upload Resource | Valid data     | Resource stored in DB     | Resource displayed in list  | Pass   |
| TC6<br>(Figure E5) | View Resource   | Click resource | Resource details shown    | Details displayed correctly | Pass   |
| TC7<br>(Figure E6) | Search          | Keyword        | Relevant results returned | Correct results displayed   | Pass   |
| TC8<br>(Figure E6) | Search          | No match       | No results message        | Message displayed           | Pass   |

Table 3: Flashcard System Testing

| Test ID             | Feature          | Input            | Expected Output               | Actual Output            | Result |
|---------------------|------------------|------------------|-------------------------------|--------------------------|--------|
| TC9<br>(Figure E8)  | Create Flashcard | Valid front/back | Flashcard saved               | Flashcard displayed      | Pass   |
| TC10<br>(Figure E7) | Link Flashcard   | Linked to source | Flashcard connected to source | Link works correctly     | Pass   |
| TC11<br>(Figure E7) | View Flashcard   | Select source    | Flashcards displayed          | Correct flashcards shown | Pass   |

Table 4: Integration Testing

| Test ID              | Feature             | Input        | Expected Output | Actual Output              | Result |
|----------------------|---------------------|--------------|-----------------|----------------------------|--------|
| TC12<br>(Figure E9)  | Source to Flashcard | Click source | Flashcards load | Flashcards displayed       | Pass   |
| TC13<br>(Figure E10) | Flashcard to Source | Click link   | Source opens    | Source displayed correctly | Pass   |
| TC14<br>(Figure E9)  | Upload to Display   | Add resource | Appears in list | Resource visible           | Pass   |

Table 5: Error Handling Testing

| Test ID              | Feature | Input             | Expected Output      | Actual Output     | Result |
|----------------------|---------|-------------------|----------------------|-------------------|--------|
| TC15<br>(Figure E2)  | Login   | Wrong credentials | Error shown          | Error displayed   | Pass   |
| TC16<br>(Figure E12) | Upload  | Missing fields    | Validation error     | Error displayed   | Pass   |
| TC17<br>(Figure E11) | Search  | Empty input       | No action or message | Handled correctly | Pass   |

## APPENDIX D - Testing Evidence

Table 1

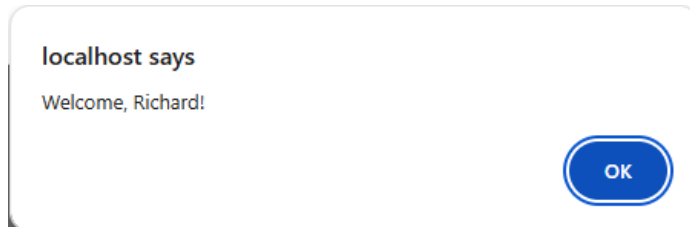


Figure E1 Successful user login

### Evidence

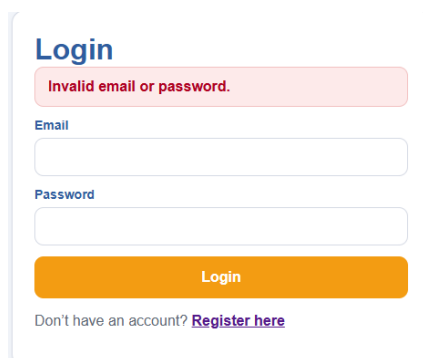


Figure E2 Error message for invalid login

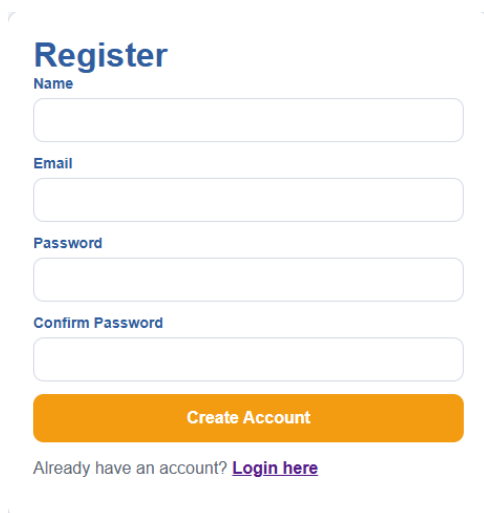


Figure E3

Table 2 Evidence

Add Source

Title

Test

Abstract

Test

Year

2026

Venue

test

Publisher

test

DOI

https://facebook.com

Website URL

https://instagram.com

PDF URL

https://instagram.com

Add Source

Figure E4 Resource successfully uploaded

PDF

Test

test (2026)

View Source

Delete Source

Figure E5 Resource successfully displayed

Search

Search across sources, flashcards, and questions in one place.

test

Search

Home

Results for: test

Sources (2)

Test

test (2026)

Test

test (2026)

Flashcards (2)

Key point from the abstract:

• test

Key point from the abstract:

• Test

Questions (0)

No questions found.

Figure E6 Search functionality returning relevant results

Table 3 Evidence

Flashcards

Review flashcards linked to this source and generate a fresh set when needed.

Back to Source

Generate Flashcards

Source: Test

Q: Key point from the abstract.

A: • Test

Save Flashcard

Generated: RULES • 2026-04-13 14:01:47

Figure E7 Flashcards linked to academic source

Create Flashcards

Creating as Richard

Module

Select a module...

Question

Answer

Save Flashcard

Figure E8 Flashcard creation interface

## Table 4 Evidence

### Spaced Repetition Promotes Efficient and Effective Learning: Policy Implications for Instruction

Nate Kornell, Robert A. Bjork — Policy Insights from the Behavioral and Brain Sciences (2008)

**Keywords:** educational psychology, learning strategies, memory retention, spaced repetition

✕

Open website

Open PDF

View Flashcards

Saved ✓

#### Abstract

This paper examines how spaced repetition improves long-term retention and learning efficiency, with implications for instructional design and educational policy.

Figure E9 Navigation from source to flashcards

### Flashcards

Review flashcards linked to this source and generate a fresh set when needed.

[← Back to Source](#) [Generate Flashcards](#)

Source: Spaced Repetition Promotes Efficient and Effective Learning: Policy Implications for Instruction

Figure E10 Navigation from flashcard to source

## Table 5 Evidence

!

Please fill in this field.

Figure E11 Validation error message for incorrect input

#### Add Source

Please provide at least a Website URL or a PDF URL.

Title

wewe

Abstract

wewe

Year

Venue

Publisher

DOI

Website URL

sdfs

PDF URL

sfd

!

Please enter a URL.

Add Source

Figure E12



## **APPENDIX E - Persona Evaluation**

### **Persona 1 – Undergraduate Student (Organisation focused user)**

This persona reflects the usual university student who manages many programs and vast amounts of academic content. The student frequently utilises Google Drive and Microsoft OneNote to save lecture notes, PDFs, and coursework resources. For this user, the system offers obvious advantages in terms of organisation and accessibility. The organised repository, together with keyword tagging and search capability, shortens the time needed to find certain items. Furthermore, the ability to build flashcards directly from saved sources avoids the need to manually transfer information into various applications, hence increasing workflow efficiency. However, this persona may still have difficulties when dealing with complicated or massive datasets, especially if sophisticated organisational features like filtering or sorting are not completely established.

### **Persona 2 - Exam focused student (Revision oriented user)**

This persona reflects a student who is primarily concerned with revision and test preparation, often using programs like Anki or Quizlet for active memory practice. This user benefits significantly from the system's integration of revision and source material. Flashcards that are closely connected to academic information allow students to return to the source material as needed, boosting contextual knowledge and lowering the danger of isolated memorisation. However, as compared to dedicated flashcard platforms, the system lacks sophisticated capabilities like performance tracking, and adaptive learning. As a result, while the system allows for modification, it may not be able to completely replace specific tools for long-term study optimisation.

### **Persona Evaluation Summary**

The persona-based evaluation indicates that the system is especially good at bridging the gap between organisation and revision, providing a more complete study experience than existing solutions. It works effectively for users who need both organised resource management and rudimentary active recall capabilities. However, the system performs less well for users who rely significantly on complex revision tools or highly customised note-taking settings. This demonstrates the trade-off between integration and feature depth, emphasising the importance of future development in improving both functionality and usability.

